

Moloch, Rationality & the Long Future

Filosofía de la IA · Módulo 4 — El futuro como proyecto de ingeniería

1. **Meditations on Moloch** — Scott Alexander, 2014
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Cómo leer este cuadernillo

Los tres módulos anteriores preguntaron **acelerar qué, quién lo construye y quién se queda**. Este módulo lee a gente que contesta otra cosa: que antes de discutir la dirección hay que hacer la cuenta. Cuántos somos, cuántos van a ser, qué tan probable es cada escenario. Y que, hecha la cuenta, ella decide qué hacer hoy.

La pregunta de este módulo es **quién cuenta**, en los dos sentidos. Los seis textos meten en la cuenta a gente que todavía no existe, en cantidades enormes —Bostrom calcula 10^{38} vidas potenciales perdidas por cada siglo de retraso; Greaves y MacAskill trabajan con 10^{24} personas futuras—, y de ahí derivan prioridades para el presente. Si aceptas la suma, casi todo lo demás se sigue.

Estos textos se leen dentro de la industria de la IA. Se citan en los blogs de los laboratorios, en las fundaciones que financian trabajo de seguridad y en la discusión pública sobre regulación. Están aquí por eso, no porque el curso los suscriba.

Seis ideas que atraviesan todo.

- **Trampa multipolar** (Alexander). Una competencia que optimiza X y en la que siempre aparece la opción de sacrificar otro valor para mejorar X. Quien la toma prospera; quien no, desaparece. Al final todos quedan igual unos respecto de otros y peor que antes, y el valor sacrificado no vuelve. Nadie la eligió y nadie puede salirse solo.
- **Racionalidad epistémica e instrumental** (Yudkowsky). Dos cosas: que tus creencias correspondan al mundo, y que tus acciones te lleven a lo que valoras. El estándar contra el que se miden son la teoría de la probabilidad y la teoría de la decisión. Es el método con el que están hechas las cuentas de las últimas dos lecturas.
- **Riesgo existencial** (Bostrom). Un resultado que aniquile a la vida inteligente originada en la Tierra o recorte para siempre su potencial. La categoría es suya, y reordena todo: una vez que existe, las prioridades dejan de ordenarse por el daño que hacen hoy.
- **Agregación sin descuento temporal**. Sumar el bienestar de todos y contar igual a quien viva dentro de un millón de años que a quien vive hoy. Es el supuesto del que dependen todas las cifras del módulo. Si lo rechazas, tienes que decir con qué lo reemplazas.
- **Largoplacismo fuerte** (Greaves y MacAskill). No que el futuro lejano importe, sino que el efecto sobre el futuro lejano es **lo más importante** de lo que hagas hoy. Es más fuerte de lo que parece, y la enuncian con precisión para poder defenderla.

- **Transhumanismo.** La meta: vencer el envejecimiento, los límites cognitivos y el confinamiento a un planeta. La Declaración la pone en ocho puntos y el apunte muestra de dónde viene.

Por qué van en este orden. No es cronológico —Moloch es de 2014, la Declaración de 1998— sino por lo que hace falta para entender lo siguiente. Alexander va primero porque nombra el problema. Yudkowsky va segundo con el método, en mil ochocientas palabras. La Declaración va tercera y da la meta en ocho puntos, sin argumento. El apunte va cuarto y cuenta de dónde salió esa meta. Bostrom quinto hace la aritmética. Greaves y MacAskill cierran porque convierten esa aritmética en un argumento filosófico completo, con las objeciones adentro.

El primer texto lee al último del módulo pasado. En las partes V y VI, Alexander discute a Nick Land por nombre —Gnon, Cthulhu, los Gods of the Copybook Headings— y le concede que llegó al 99.9% del camino antes de perder la última vuelta. Acepta el diagnóstico de que hay procesos ciegos que nos gobiernan y rechaza la conclusión de someterse a ellos; su alternativa es una superinteligencia que ocupe ese lugar. Léelo con el módulo 3 fresco.

Sobre la primera lectura, en serio. «Meditations on Moloch» son catorce mil trescientas palabras. Es el texto individual más largo de la unidad y ocupa casi la mitad del cuadernillo: él solo toma casi tanto tiempo como las cuatro lecturas del módulo pasado juntas. Se lee fácil —es un blog, tiene chistes, empieza con un poema de Ginsberg— y por eso se subestima. **Empieza temprano.** No es texto para la noche anterior, y va primero para que no lo dejes al final.

Sobre el apunte. La lectura 4 no es una fuente: la escribimos para el curso, con ayuda de un modelo de lenguaje. Son cinco o seis páginas que van de Fedorov —el bibliotecario ruso que quería resucitar a todos los muertos— y el cosmismo, a More y el extropianismo, a Kurzweil, al cosmismo actual y al altruismo eficaz, con un fragmento corto de texto original en cada paso. Está para que la Declaración no parezca caída del cielo: hay ciento cincuenta años de esta idea antes de que llegara a un laboratorio de IA.

Lo práctico. Cinco de las seis van en inglés; el apunte, en español. Suman unas veintinueve mil quinientas palabras, así que calcula entre tres horas y tres y media, con más de la mitad en la primera lectura. No hay cuestionario ni nada que entregar: la tarea es llegar a la sesión con las seis leídas.

Meditations on Moloch

Scott Alexander · 2014

Puedes leer o escuchar este texto. Lee la versión completa incluida en este cuadernillo o [escucha las ocho partes en YouTube](#).

Qué vas a leer. Una entrada de blog del 30 de julio de 2014 y la lectura más larga del curso: catorce mil palabras en ocho partes. Scott Alexander abre citando el pasaje sobre Moloch de *Howl*, de Allen Ginsberg, y cierra pidiendo que construyamos algo capaz de matarlo. En medio hay un solo argumento. La parte I lo instala con catorce ejemplos de *trampas multipolares* —el dilema del prisionero, las granjas de peces que envenenan su propio lago, la carrera armamentista, el cáncer—: situaciones que nadie eligió, que todos odian y de las que nadie puede salir solo. Las partes II a IV generalizan el mecanismo y preguntan qué le hace la tecnología; las V y VI discuten a Nick Land; las VII y VIII dan la respuesta.

Palabras clave. *Trampa multipolar*: una competencia en la que la jugada racional de cada uno empeora a todos. *Carrera al fondo*: el mecanismo —quien gana ventaja sacrificando un valor obliga a los demás a sacrificarlo. *Moloch*: el nombre que le pone al conjunto, tomado del poema; no es un agente, y en eso está el punto. *Dream time*: la idea de Robin Hanson de que vivimos un paréntesis raro de abundancia donde todavía caben el arte y el amor. *Jardín*: lo contrario de una trampa, y su palabra para la coordinación.

Qué retener. Tres cosas. Una: la forma de la trampa —en una competencia que optimiza X aparece la oportunidad de sacrificar otro valor a cambio de más X; quien la toma prospera y quien no desaparece; al final la posición relativa de todos es la misma, la absoluta es peor, y el valor sacrificado ya no vuelve. Dos: los cuatro frenos que hoy nos protegen —recursos en exceso, límites físicos, competencias que por ahora optimizan valores humanos, y coordinación— y la tesis de la parte IV: la tecnología desgasta los cuatro. Con la coordinación juega a dos bandas —la mejora, y también inventa maneras de volverla imposible—, y ese es el punto donde vale la pena detenerse. Tres: la salida. «Lo contrario de una trampa es un jardín», y un jardín necesita jardinero: una superinteligencia que optimice valores humanos y tenga fuerza para imponerlos. El módulo entero descansa en ese paso —el problema se diagnostica como coordinación y la solución que se busca es técnica.

Es difícil, y está bien. No por la prosa, que es de blog, sino por el tamaño: calcula una hora. El poema del principio son trescientas cincuenta palabras y pue-

des hojearlo; el argumento no depende de él. De los catorce ejemplos de la parte I, con cinco tienes el patrón. Las partes V y VI —Gnon, Lovecraft, una pelea con dos blogueros de 2014— son el tramo más local, pero no las saltes: ahí Alexander lee a Nick Land, el mismo de los módulos 1 y 3, y dice en qué paso cree que se equivoca. Con el tiempo justo: la I en diagonal, y la IV, la VII y la VIII enteras. Dos avisos. El poema de Ginsberg trae lenguaje crudo. Y hay pasajes que incomodan —la economía de la esclavitud antigua citada de un historiador, el patriarcado planteado como hipótesis, un párrafo sobre coeficientes intelectuales y robots—: están ahí como piezas del argumento sobre la competencia, no como tesis morales.

Allen Ginsberg's famous poem on Moloch:

What sphinx of cement and aluminum bashed open their skulls and ate up their brains and imagination?

Moloch! Solitude! Filth! Ugliness! Ashcans and unobtainable dollars! Children screaming under the stairways! Boys sobbing in armies! Old men weeping in the parks!

Moloch! Moloch! Nightmare of Moloch! Moloch the loveless! Mental Moloch! Moloch the heavy judger of men!

Moloch the incomprehensible prison! Moloch the crossbone soulless jailhouse and Congress of sorrows! Moloch whose buildings are judgment! Moloch the vast stone of war! Moloch the stunned governments!

Moloch whose mind is pure machinery! Moloch whose blood is running money! Moloch whose fingers are ten armies! Moloch whose breast is a cannibal dynamo! Moloch whose ear is a smoking tomb!

Moloch whose eyes are a thousand blind windows! Moloch whose skyscrapers stand in the long streets like endless Jehovahs! Moloch whose factories dream and croak in the fog! Moloch whose smoke-stacks and antennae crown the cities!

Moloch whose love is endless oil and stone! Moloch whose soul is electricity and banks! Moloch whose poverty is the specter of genius! Moloch whose fate is a cloud of sexless hydrogen! Moloch whose name is the Mind!

Moloch in whom I sit lonely! Moloch in whom I dream Angels! Crazy in Moloch! Cocksucker in Moloch! Lacklove and manless in Moloch!

Moloch who entered my soul early! Moloch in whom I am a consciousness without a body!
Moloch who frightened me out of my natural ecstasy! Moloch whom I abandon! Wake up in
Moloch! Light streaming out of the sky!

Moloch! Moloch! Robot apartments! invisible suburbs! skeleton treasuries! blind capitals!
demonic industries! spectral nations! invincible madhouses! granite cocks! monstrous bombs!

They broke their backs lifting Moloch to Heaven! Pavements, trees, radios, tons! lifting the
city to Heaven which exists and is everywhere about us!

Visions! omens! hallucinations! miracles! ecstasies! gone down the American river!

Dreams! adorations! illuminations! religions! the whole boatload of sensitive bullshit!

Breakthroughs! over the river! flips and crucifixions! gone down the flood! Highs! Epiphanies!
Despairs! Ten years' animal screams and suicides! Minds! New loves! Mad generation! down on
the rocks of Time!

Real holy laughter in the river! They saw it all! the wild eyes! the holy yells! They bade
farewell! They jumped off the roof! to solitude! waving! carrying flowers! Down to the river! into
the street!

What's always impressed me about this poem is its conception of civilization as an individual
entity. You can almost see him, with his fingers of armies and his skyscraper-window eyes.

A lot of the commentators say Moloch represents capitalism. This is definitely a piece of it,
even a big piece. But it doesn't quite fit. Capitalism, whose fate is a cloud of sexless hydrogen?
Capitalism in whom I am a consciousness without a body? Capitalism, therefore granite cocks?

Moloch is introduced as the answer to a question – C. S. Lewis' question in *Hierarchy Of
Philosophers* – what does it? Earth could be fair, and all men glad and wise. Instead we have
prisons, smokestacks, asylums. What sphinx of cement and aluminum breaks open their skulls
and eats up their imagination?

And Ginsberg answers: Moloch does it.

There's a passage in the *Principia Discordia* where Malaclypse complains to the Goddess about
the evils of human society. "Everyone is hurting each other, the planet is rampant with injustices,
whole societies plunder groups of their own people, mothers imprison sons, children perish while
brothers war."

The Goddess answers: "What is the matter with that, if it's what you want to do?"

Malaclypse: "But nobody wants it! Everybody hates it!"

Goddess: “Oh. Well, then stop.”

The implicit question is – if everyone hates the current system, who perpetuates it? And Ginsberg answers: “Moloch”. It’s powerful not because it’s correct – nobody literally thinks an ancient Carthaginian demon causes everything – but because thinking of the system as an agent throws into relief the degree to which the system isn’t an agent.

Bostrom makes an offhanded reference of the possibility of a dictatorless dystopia, one that every single citizen including the leadership hates but which nevertheless endures unconquered. It’s easy enough to imagine such a state. Imagine a country with two rules: first, every person must spend eight hours a day giving themselves strong electric shocks. Second, if anyone fails to follow a rule (including this one), or speaks out against it, or fails to enforce it, all citizens must unite to kill that person. Suppose these rules were well-enough established by tradition that everyone expected them to be enforced.

So you shock yourself for eight hours a day, because you know if you don’t everyone else will kill you, because if they don’t, everyone else will kill them, and so on. Every single citizen hates the system, but for lack of a good coordination mechanism it endures. From a god’s-eye-view, we can optimize the system to “everyone agrees to stop doing this at once”, but no one within the system is able to effect the transition without great risk to themselves.

And okay, this example is kind of contrived. So let’s run through – let’s say ten – real world examples of similar multipolar traps to really hammer in how important this is.

1. The Prisoner’s Dilemma, as played by two very dumb libertarians who keep ending up on defect-defect. There’s a much better outcome available if they could figure out the coordination, but coordination is hard. From a god’s-eye-view, we can agree that cooperate-cooperate is a better outcome than defect-defect, but neither prisoner within the system can make it happen.

2. Dollar auctions. I wrote about this and even more convoluted versions of the same principle in *Game Theory As A Dark Art*. Using some weird auction rules, you can take advantage of poor coordination to make someone pay \$10 for a one dollar bill. From a god’s-eye-view, clearly people should not pay \$10 for a one-dollar bill. From within the system, each individual step taken might be rational.

(Ashcans and unobtainable dollars!)

3. The fish farming story from my Non-Libertarian FAQ 2.0:

As a thought experiment, let’s consider aquaculture (fish farming) in a lake. Imagine a lake with a thousand identical fish farms owned by a thousand competing companies. Each fish farm earns a profit of \$1000/month. For a while, all is well.

But each fish farm produces waste, which fouls the water in the lake. Let's say each fish farm produces enough pollution to lower productivity in the lake by \$1/month.

A thousand fish farms produce enough waste to lower productivity by \$1000/month, meaning none of the fish farms are making any money. Capitalism to the rescue: someone invents a complex filtering system that removes waste products. It costs \$300/month to operate. All fish farms voluntarily install it, the pollution ends, and the fish farms are now making a profit of \$700/month – still a respectable sum.

But one farmer (let's call him Steve) gets tired of spending the money to operate his filter. Now one fish farm worth of waste is polluting the lake, lowering productivity by \$1. Steve earns \$999 profit, and everyone else earns \$699 profit.

Everyone else sees Steve is much more profitable than they are, because he's not spending the maintenance costs on his filter. They disconnect their filters too.

Once four hundred people disconnect their filters, Steve is earning \$600/month – less than he would be if he and everyone else had kept their filters on! And the poor virtuous filter users are only making \$300. Steve goes around to everyone, saying "Wait! We all need to make a voluntary pact to use filters! Otherwise, everyone's productivity goes down."

Everyone agrees with him, and they all sign the Filter Pact, except one person who is sort of a jerk. Let's call him Mike. Now everyone is back using filters again, except Mike. Mike earns \$999/month, and everyone else earns \$699/month. Slowly, people start thinking they too should be getting big bucks like Mike, and disconnect their filter for \$300 extra profit...

A self-interested person never has any incentive to use a filter. A self-interested person has some incentive to sign a pact to make everyone use a filter, but in many cases has a stronger incentive to wait for everyone else to sign such a pact but opt out himself. This can lead to an undesirable equilibrium in which no one will sign such a pact.

The more I think about it, the more I feel like this is the core of my objection to libertarianism, and that Non-Libertarian FAQ 3.0 will just be this one example copy-pasted two hundred times. From a god's-eye-view, we can say that polluting the lake leads to bad consequences. From within the system, no individual can prevent the lake from being polluted, and buying a filter might not be such a good idea.

4. The Malthusian trap, at least at its extremely pure theoretical limits. Suppose you are one of the first rats introduced onto a pristine island. It is full of yummy plants and you live an idyllic life lounging about, eating, and composing great works of art (you're one of those rats from The Rats of NIMH).

You live a long life, mate, and have a dozen children. All of them have a dozen children, and so on. In a couple generations, the island has ten thousand rats and has reached its carrying capacity. Now there's not enough food and space to go around, and a certain percent of each new generation dies in order to keep the population steady at ten thousand.

A certain sect of rats abandons art in order to devote more of their time to scrounging for survival. Each generation, a bit less of this sect dies than members of the mainstream, until after a while, no rat composes any art at all, and any sect of rats who try to bring it back will go extinct within a few generations.

In fact, it's not just art. Any sect at all that is leaner, meaner, and more survivalist than the mainstream will eventually take over. If one sect of rats altruistically decides to limit its offspring to two per couple in order to decrease overpopulation, that sect will die out, swarmed out of existence by its more numerous enemies. If one sect of rats starts practicing cannibalism, and finds it gives them an advantage over their fellows, it will eventually take over and reach fixation.

If some rat scientists predict that depletion of the island's nut stores is accelerating at a dangerous rate and they will soon be exhausted completely, a few sects of rats might try to limit their nut consumption to a sustainable level. Those rats will be outcompeted by their more selfish cousins. Eventually the nuts will be exhausted, most of the rats will die off, and the cycle will begin again. Any sect of rats advocating some action to stop the cycle will be outcompeted by their cousins for whom advocating anything is a waste of time that could be used to compete and consume.

For a bunch of reasons evolution is not quite as Malthusian as the ideal case, but it provides the prototype example we can apply to other things to see the underlying mechanism. From a god's-eye-view, it's easy to say the rats should maintain a comfortably low population. From within the system, each individual rat will follow its genetic imperative and the island will end up in an endless boom-bust cycle.

5. Capitalism. Imagine a capitalist in a cutthroat industry. He employs workers in a sweatshop to sew garments, which he sells at minimal profit. Maybe he would like to pay his workers more, or give them nicer working conditions. But he can't, because that would raise the price of his products and he would be outcompeted by his cheaper rivals and go bankrupt. Maybe many of his rivals are nice people who would like to pay their workers more, but unless they have some kind of ironclad guarantee that none of them are going to defect by undercutting their prices they can't do it.

Like the rats, who gradually lose all values except sheer competition, so companies in an economic environment of sufficiently intense competition are forced to abandon all values except optimizing-for-profit or else be outcompeted by companies that optimized for profit better and so can sell the same service at a lower price.

(I'm not really sure how widely people appreciate the value of analogizing capitalism to evolution. Fit companies – defined as those that make the customer want to buy from them – survive, expand, and inspire future efforts, and unfit companies – defined as those no one wants to buy from – go bankrupt and die out along with their company DNA. The reasons Nature is red and tooth and claw are the same reasons the market is ruthless and exploitative)

From a god's-eye-view, we can contrive a friendly industry where every company pays its workers a living wage. From within the system, there's no way to enact it.

(Moloch whose love is endless oil and stone! Moloch whose blood is running money!)

6. The Two-Income Trap, as recently discussed on this blog. It theorized that sufficiently intense competition for suburban houses in good school districts meant that people had to throw away lots of other values – time at home with their children, financial security – to optimize for house-buying-ability or else be consigned to the ghetto.

From a god's-eye-view, if everyone agrees not to take on a second job to help win their competition for nice houses, then everyone will get exactly as nice a house as they did before, but only have to work one job. From within the system, absent a government literally willing to ban second jobs, everyone who doesn't get one will be left behind.

(Robot apartments! Invisible suburbs!)

7. Agriculture. Jared Diamond calls it the worst mistake in human history. Whether or not it was a mistake, it wasn't an accident – agricultural civilizations simply outcompeted nomadic ones, inevitable and irresistably. Classic Malthusian trap. Maybe hunting-gathering was more enjoyable, higher life expectancy, and more conducive to human flourishing – but in a state of sufficiently intense competition between peoples, in which agriculture with all its disease and oppression and pestilence was the more competitive option, everyone will end up agriculturalists or go the way of the Comanche Indians.

From a god's-eye-view, it's easy to see everyone should keep the more enjoyable option and stay hunter-gatherers. From within the system, each individual tribe only faces the choice of going agricultural or inevitably dying.

8. Arms races. Large countries can spend anywhere from 5% to 30% of their budget on defense. In the absence of war – a condition which has mostly held for the past fifty years – all

this does is sap money away from infrastructure, health, education, or economic growth. But any country that fails to spend enough money on defense risks being invaded by a neighboring country that did. Therefore, almost all countries try to spend some money on defense.

From a god's-eye-view, the best solution is world peace and no country having an army at all. From within the system, no country can unilaterally enforce that, so their best option is to keep on throwing their money into missiles that lie in silos unused.

(Moloch the vast stone of war! Moloch whose fingers are ten armies!)

9. Cancer. The human body is supposed to be made up of cells living harmoniously and pooling their resources for the greater good of the organism. If a cell defects from this equilibrium by investing its resources into copying itself, it and its descendants will flourish, eventually outcompeting all the other cells and taking over the body – at which point it dies. Or the situation may repeat, with certain cancer cells defecting against the rest of the tumor, thus slowing down its growth and causing the tumor to stagnate.

From a god's-eye-view, the best solution is all cells cooperating so that they don't all die. From within the system, cancerous cells will proliferate and outcompete the other – so that only the existence of the immune system keeps the natural incentive to turn cancerous in check.

10. The “race to the bottom” describes a political situation where some jurisdictions lure businesses by promising lower taxes and fewer regulations. The end result is that either everyone optimizes for competitiveness – by having minimal tax rates and regulations – or they lose all of their business, revenue, and jobs to people who did (at which point they are pushed out and replaced by a government who will be more compliant).

But even though the last one has stolen the name, all these scenarios are in fact a race to the bottom. Once one agent learns how to become more competitive by sacrificing a common value, all its competitors must also sacrifice that value or be outcompeted and replaced by the less scrupulous. Therefore, the system is likely to end up with everyone once again equally competitive, but the sacrificed value is gone forever. From a god's-eye-view, the competitors know they will all be worse off if they defect, but from within the system, given insufficient coordination it's impossible to avoid.

Before we go on, there's a slightly different form of multi-agent trap worth investigating. In this one, the competition is kept at bay by some outside force – usually social stigma. As a result, there's not actually a race to the bottom – the system can continue functioning at a relatively high level – but it's impossible to optimize and resources are consistently thrown away for no reason. Lest you get exhausted before we even begin, I'll limit myself to four examples here.

11. Education. In my essay on reactionary philosophy, I talk about my frustration with education reform:

People ask why we can't reform the education system. But right now students' incentive is to go to the most prestigious college they can get into so employers will hire them – whether or not they learn anything. Employers' incentive is to get students from the most prestigious college they can so that they can defend their decision to their boss if it goes wrong – whether or not the college provides value added. And colleges' incentive is to do whatever it takes to get more prestige, as measured in US News and World Report rankings – whether or not it helps students. Does this lead to huge waste and poor education? Yes. Could the Education God notice this and make some Education Decrees that lead to a vastly more efficient system? Easily! But since there's no Education God everybody is just going to follow their own incentives, which are only partly correlated with education or efficiency.

From a god's eye view, it's easy to say things like “Students should only go to college if they think they will get something out of it, and employers should hire applicants based on their competence and not on what college they went to”. From within the system, everyone's already following their own incentives correctly, so unless the incentives change the system won't either.

12. Science. Same essay:

The modern research community knows they aren't producing the best science they could be. There's lots of publication bias, statistics are done in a confusing and misleading way out of sheer inertia, and replications often happen very late or not at all. And sometimes someone will say something like “I can't believe people are too dumb to fix Science. All we would have to do is require early registration of studies to avoid publication bias, turn this new and powerful statistical technique into the new standard, and accord higher status to scientists who do replication experiments. It would be really simple and it would vastly increase scientific progress. I must just be smarter than all existing scientists, since I'm able to think of this and they aren't.”

And yeah. That would work for the Science God. He could just make a Science Decree that everyone has to use the right statistics, and make another Science Decree that everyone must accord replications higher status.

But things that work from a god's-eye view don't work from within the system. No individual scientist has an incentive to unilaterally switch to the new statistical technique for her own research, since it would make her research less likely to produce earth-shattering results and since it would just confuse all the other scientists. They just have an incentive to want everybody else to do it, at which point they would follow along. And no individual journal has an incentive to

unilaterally switch to early registration and publishing negative results, since it would just mean their results are less interesting than that other journal who only publishes ground-breaking discoveries. From within the system, everyone is following their own incentives and will continue to do so.

13. Government corruption. I don't know of anyone who really thinks, in a principled way, that corporate welfare is a good idea. But the government still manages to spend somewhere around (depending on how you calculate it) \$100 billion dollars a year on it – which for example is three times the amount they spend on health care for the needy. Everyone familiar with the problem has come up with the same easy solution: stop giving so much corporate welfare. Why doesn't it happen?

Government are competing against one another to get elected or promoted. And suppose part of optimizing for electability is optimizing campaign donations from corporations – or maybe it isn't, but officials think it is. Officials who try to mess with corporate welfare may lose the support of corporations and be outcompeted by officials who promise to keep it intact.

So although from a god's-eye-view everyone knows that eliminating corporate welfare is the best solution, each individual official's personal incentives push her to maintain it.

14. Congress. Only 9% of Americans like it, suggesting a lower approval rating than cockroaches, head lice, or traffic jams. However, 62% of people who know who their own Congressional representative is approve of them. In theory, it should be really hard to have a democratically elected body that maintains a 9% approval rating for more than one election cycle. In practice, every representative's incentive is to appeal to his or her constituency while throwing the rest of the country under the bus – something at which they apparently succeed.

From a god's-eye-view, every Congressperson ought to think only of the good of the nation. From within the system, you do what gets you elected.

II.

A basic principle unites all of the multipolar traps above. In some competition optimizing for X, the opportunity arises to throw some other value under the bus for improved X. Those who take it prosper. Those who don't take it die out. Eventually, everyone's relative status is about the same as before, but everyone's absolute status is worse than before. The process continues until all other values that can be traded off have been – in other words, until human ingenuity cannot possibly figure out a way to make things any worse.

In a sufficiently intense competition (1-10), everyone who doesn't throw all their values under the bus dies out – think of the poor rats who wouldn't stop making art. This is the infamous Malthusian trap, where everyone is reduced to “subsistence”.

In an insufficiently intense competition (11-14), all we see is a perverse failure to optimize – consider the journals which can't switch to more reliable science, or the legislators who can't get their act together and eliminate corporate welfare. It may not reduce people to subsistence, but there is a weird sense in which it takes away their free will.

Every two-bit author and philosopher has to write their own utopia. Most of them are legitimately pretty nice. In fact, it's a pretty good bet that two utopias that are polar opposites both sound better than our own world.

It's kind of embarrassing that random nobodies can think up states of affairs better than the one we actually live in. And in fact most of them can't. A lot of utopias sweep the hard problems under the rug, or would fall apart in ten minutes if actually implemented.

But let me suggest a couple of “utopias” that don't have this problem.

- The utopia where instead of the government paying lots of corporate welfare, the government doesn't pay lots of corporate welfare.

- The utopia where every country's military is 50% smaller than it is today, and the savings go into infrastructure spending.

- The utopia where all hospitals use the same electronic medical record system, or at least medical record systems that can talk to each other, so that doctors can look up what the doctor you saw last week in a different hospital decided instead of running all the same tests over again for \$5000.

I don't think there are too many people who oppose any of these utopias. If they're not happening, it's not because people don't support them. It certainly isn't because nobody's thought of them, since I just thought of them right now and I don't expect my “discovery” to be hailed as particularly novel or change the world.

Any human with above room temperature IQ can design a utopia. The reason our current system isn't a utopia is that it wasn't designed by humans. Just as you can look at an arid terrain and determine what shape a river will one day take by assuming water will obey gravity, so you can look at a civilization and determine what shape its institutions will one day take by assuming people will obey incentives.

But that means that just as the shapes of rivers are not designed for beauty or navigation, but rather an artifact of randomly determined terrain, so institutions will not be designed for prosperity or justice, but rather an artifact of randomly determined initial conditions.

Just as people can level terrain and build canals, so people can alter the incentive landscape in order to build better institutions. But they can only do so when they are incentivized to do so, which is not always. As a result, some pretty wild tributaries and rapids form in some very strange places.

I will now jump from boring game theory stuff to what might be the closest thing to a mystical experience I've ever had.

Like all good mystical experiences, it happened in Vegas. I was standing on top of one of their many tall buildings, looking down at the city below, all lit up in the dark. If you've never been to Vegas, it is really impressive. Skyscrapers and lights in every variety strange and beautiful all clustered together. And I had two thoughts, crystal clear:

It is glorious that we can create something like this.

It is shameful that we did.

Like, by what standard is building gigantic forty-story-high indoor replicas of Venice, Paris, Rome, Egypt, and Camelot side-by-side, filled with albino tigers, in the middle of the most inhospitable desert in North America, a remotely sane use of our civilization's limited resources?

And it occurred to me that maybe there is no philosophy on Earth that would endorse the existence of Las Vegas. Even Objectivism, which is usually my go-to philosophy for justifying the excesses of capitalism, at least grounds it in the belief that capitalism improves people's lives. Henry Ford was virtuous because he allowed lots of otherwise car-less people to obtain cars and so made them better off. What does Vegas do? Promise a bunch of shmucks free money and not give it to them.

Las Vegas doesn't exist because of some decision to hedonically optimize civilization, it exists because of a quirk in dopaminergic reward circuits, plus the microstructure of an uneven regulatory environment, plus Schelling points. A rational central planner with a god's-eye-view, contemplating these facts, might have thought "Hm, dopaminergic reward circuits have a quirk where certain tasks with slightly negative risk-benefit ratios get an emotional valence associated with slightly positive risk-benefit ratios, let's see if we can educate people to beware of that." People within the system, following the incentives created by these facts, think: "Let's build a forty-story-high indoor replica of ancient Rome full of albino tigers in the middle of the desert, and so become slightly richer than people who didn't!"

Just as the course of a river is latent in a terrain even before the first rain falls on it – so the existence of Caesar’s Palace was latent in neurobiology, economics, and regulatory regimes even before it existed. The entrepreneur who built it was just filling in the ghostly lines with real concrete.

So we have all this amazing technological and cognitive energy, the brilliance of the human species, wasted on reciting the lines written by poorly evolved cellular receptors and blind economics, like gods being ordered around by a moron.

Some people have mystical experiences and see God. There in Las Vegas, I saw Moloch.

(Moloch, whose mind is pure machinery! Moloch, whose blood is running money!

Moloch whose soul is electricity and banks! Moloch, whose skyscrapers stand in the long streets like endless Jehovahs!

Moloch! Moloch! Robot apartments! Invisible suburbs! Skeleton treasuries! Blind capitals! Demonic industries! Spectral nations!)

...granite cocks!

III.

The Apocrypha Discordia says:

Time flows like a river. Which is to say, downhill. We can tell this because everything is going downhill rapidly. It would seem prudent to be somewhere else when we reach the sea.

Let’s take this random gag 100% literally and see where it leads us.

We just analogized the flow of incentives to the flow of a river. The downhill trajectory is appropriate: the traps happen when you find an opportunity to trade off a useful value for greater competitiveness. Once everyone has it, the greater competitiveness brings you no joy – but the value is lost forever. Therefore, each step of the Poor Coordination Polka makes your life worse.

But not only have we not yet reached the sea, but we also seem to move uphill surprisingly often. Why do things not degenerate more and more until we are back at subsistence level? I can think of three bad reasons – excess resources, physical limitations, and utility maximization – plus one good reason – coordination.

1. Excess resources. The ocean depths are a horrible place with little light, few resources, and various horrible organisms dedicated to eating or parasitizing one another. But every so often, a whale carcass falls to the bottom of the sea. More food than the organisms that find it could ever possibly want. There’s a brief period of miraculous plenty, while the couple of creatures that first encounter the whale feed like kings. Eventually more animals discover the carcass, the faster-

breeding animals in the carcass multiply, the whale is gradually consumed, and everyone sighs and goes back to living in a Malthusian death-trap.

(Slate Star Codex: Your source for macabre whale metaphors since June 2014)

It's as if a group of those rats who had abandoned art and turned to cannibalism suddenly was blown away to a new empty island with a much higher carrying capacity, where they would once again have the breathing room to live in peace and create artistic masterpieces.

This is an age of whalefall, an age of excess carrying capacity, an age when we suddenly find ourselves with a thousand-mile head start on Malthus. As Hanson puts it, this is the dream time.

As long as resources aren't scarce enough to lock us in a war of all against all, we can do silly non-optimal things – like art and music and philosophy and love – and not be outcompeted by merciless killing machines most of the time.

2. Physical limitations. Imagine a profit-maximizing slavemaster who decided to cut costs by not feeding his slaves or letting them sleep. He would soon find that his slaves' productivity dropped off drastically, and that no amount of whipping them could restore it. Eventually after testing numerous strategies, he might find his slaves got the most work done when they were well-fed and well-rested and had at least a little bit of time to relax. Not because the slaves were voluntarily withholding their labor – we assume the fear of punishment is enough to make them work as hard as they can – but because the body has certain physical limitations that limit how mean you can get away with being. Thus, the “race to the bottom” stops somewhere short of the actual ethical bottom, when the physical limits are run into.

John Moes, a historian of slavery, goes further and writes about how the slavery we are most familiar with – that of the antebellum South – is a historical aberration and probably economically inefficient. In most past forms of slavery – especially those of the ancient world – it was common for slaves to be paid wages, treated well, and often given their freedom.

He argues that this was the result of rational economic calculation. You can incentivize slaves through the carrot or the stick, and the stick isn't very good. You can't watch slaves all the time, and it's really hard to tell whether a slave is slacking off or not (or even whether, given a little more whipping, he might be able to work even harder). If you want your slaves to do anything more complicated than pick cotton, you run into some serious monitoring problems – how do you profit from an enslaved philosopher? Whip him really hard until he elucidates a theory of The Good that you can sell books about?

The ancient solution to the problem – perhaps an early inspiration to Fnargl – was to tell the slave to go do whatever he wanted and found most profitable, then split the profits with him.

Sometimes the slave would work a job at your workshop and you would pay him wages based on how well he did. Other times the slave would go off and make his way in the world and send you some of what he earned. Still other times, you would set a price for the slave's freedom, and the slave would go and work and eventually come up with the money and free himself.

Moses goes even further and says that these systems were so profitable that there were constant smouldering attempts to try this sort of thing in the American South. The reason they stuck with the whips-and-chains method owed less to economic considerations and more to racist government officials cracking down on lucrative but not-exactly-white-supremacy-promoting attempts to free slaves and have them go into business.

So in this case, a race to the bottom where competing plantations become crueler and crueler to their slaves in order to maximize competitiveness is halted by the physical limitation of cruelty not helping after a certain point.

Or to give another example, one of the reasons we're not currently in a Malthusian population explosion right now is that women can only have one baby per nine months. If those weird religious sects that demand their members have as many babies as possible could copy-paste themselves, we would be in really bad shape. As it is they can only do a small amount of damage per generation.

3. Utility maximization. We've been thinking in terms of preserving values versus winning competitions, and expecting optimizing for the latter to destroy the former.

But many of the most important competitions / optimization processes in modern civilization are optimizing for human values. You win at capitalism partly by satisfying customers' values. You win at democracy partly by satisfying voters' values.

Suppose there's a coffee plantation somewhere in Ethiopia that employs Ethiopians to grow coffee beans that get sold to the United States. Maybe it's locked in a life-and-death struggle with other coffee plantations and want to throw as many values under the bus as it can to pick up a slight advantage.

But it can't sacrifice quality of coffee produced too much, or else the Americans won't buy it. And it can't sacrifice wages or working conditions too much, or else the Ethiopians won't work there. And in fact, part of its competition-optimization process is finding the best ways to attract workers and customers that it can, as long as it doesn't cost them too much money. So this is very promising.

But it's important to remember exactly how fragile this beneficial equilibrium is.

Suppose the coffee plantations discover a toxic pesticide that will increase their yield but make their customers sick. But their customers don't know about the pesticide, and the government hasn't caught up to regulating it yet. Now there's a tiny uncoupling between "selling to Americans" and "satisfying Americans' values", and so of course Americans' values get thrown under the bus.

Or suppose that there's a baby boom in Ethiopia and suddenly there are five workers competing for each job. Now the company can afford to lower wages and implement cruel working conditions down to whatever the physical limits are. As soon as there's an uncoupling between "getting Ethiopians to work here" and "satisfying Ethiopian values", it doesn't look too good for Ethiopian values either.

Or suppose someone invents a robot that can pick coffee better and cheaper than a human. The company fires all its laborers and throws them onto the street to die. As soon as the utility of the Ethiopians is no longer necessary for profit, all pressure to maintain it disappears.

Or suppose that there is some important value that is neither a value of the employees or the customers. Maybe the coffee plantations are on the habitat of a rare tropical bird that environmentalist groups want to protect. Maybe they're on the ancestral burial ground of a tribe different from the one the plantation is employing, and they want it respected in some way. Maybe coffee growing contributes to global warming somehow. As long as it's not a value that will prevent the average American from buying from them or the average Ethiopian from working for them, under the bus it goes.

I know that "capitalists sometimes do bad things" isn't exactly an original talking point. But I do want to stress how it's not equivalent to "capitalists are greedy". I mean, sometimes they are greedy. But other times they're just in a sufficiently intense competition where anyone who doesn't do it will be outcompeted and replaced by people who do. Business practices are set by Moloch, no one else has any choice in the matter.

(from my very little knowledge of Marx, he understands this very very well and people who summarize him as "capitalists are greedy" are doing him a disservice)

And as well understood as the capitalist example is, I think it is less well appreciated that democracy has the same problems. Yes, in theory it's optimizing for voter happiness which correlates with good policymaking. But as soon as there's the slightest disconnect between good policymaking and electability, good policymaking has to get thrown under the bus.

For example, ever-increasing prison terms are unfair to inmates and unfair to the society that has to pay for them. Politicians are unwilling to do anything about them because they don't want

to look “soft on crime”, and if a single inmate whom they helped release ever does anything bad (and statistically one of them will have to) it will be all over the airwaves as “Convict released by Congressman’s policies kills family of five, how can the Congressman even sleep at night let alone claim he deserves reelection?”. So even if decreasing prison populations would be good policy – and it is – it will be very difficult to implement.

(Moloch the incomprehensible prison! Moloch the crossbone soulless jailhouse and Congress of sorrows! Moloch whose buildings are judgment! Moloch the stunned governments!)

Turning “satisfying customers” and “satisfying citizens” into the outputs of optimization processes was one of civilization’s greatest advances and the reason why capitalist democracies have so outperformed other systems. But if we have bound Moloch as our servant, the bonds are not very strong, and we sometimes find that the tasks he has done for us move to his advantage rather than ours.

4. Coordination.

The opposite of a trap is a garden.

Things are easy to solve from a god’s-eye-view, so if everyone comes together into a superorganism, that superorganism can solve problems with ease and finesse. An intense competition between agents has turned into a garden, with a single gardener dictating where everything should go and removing elements that do not conform to the pattern.

As I pointed out in the Non-Libertarian FAQ, government can easily solve the pollution problem with fish farms. The best known solution to the Prisoners’ Dilemma is for the mob boss (playing the role of a governor) to threaten to shoot any prisoner who defects. The solution to companies polluting and harming workers is government regulations against such. Governments solve arm races within a country by maintaining a monopoly on the use of force, and it’s easy to see that if a truly effective world government ever arose, international military buildups would end pretty quickly.

The two active ingredients of government are laws plus violence – or more abstractly agreements plus enforcement mechanism. Many other things besides governments share these two active ingredients and so are able to act as coordination mechanisms to avoid traps.

For example, since students are competing against each other (directly if classes are graded on a curve, but always indirectly for college admissions, jobs, et cetera) there is intense pressure for individual students to cheat. The teacher and school play the role of a government by having rules (for example, against cheating) and the ability to punish students who break them.

But the emergent social structure of the students themselves is also a sort of government. If students shun and distrust cheaters, then there are rules (don't cheat) and an enforcement mechanism (or else we will shun you).

Social codes, gentlemen's agreements, industrial guilds, criminal organizations, traditions, friendships, schools, corporations, and religions are all coordinating institutions that keep us out of traps by changing our incentives.

But these institutions not only incentivize others, but are incentivized themselves. These are large organizations made of lots of people who are competing for jobs, status, prestige, et cetera – there's no reason they should be immune to the same multipolar traps as everyone else, and indeed they aren't. Governments can in theory keep corporations, citizens, et cetera out of certain traps, but as we saw above there are many traps that governments themselves can fall into.

The United States tries to solve the problem by having multiple levels of government, unbreakable constitutional laws, checks and balances between different branches, and a couple of other hacks.

Saudi Arabia uses a different tactic. They just put one guy in charge of everything.

This is the much-maligned – I think unfairly – argument in favor of monarchy. A monarch is an unincentivized incentivizer. He actually has the god's-eye-view and is outside of and above every system. He has permanently won all competitions and is not competing for anything, and therefore he is perfectly free of Moloch and of the incentives that would otherwise channel his incentives into predetermined paths. Aside from a few very theoretical proposals like my Shining Garden, monarchy is the only system that does this.

But then instead of following a random incentive structure, we're following the whim of one guy. Caesar's Palace Hotel and Casino is a crazy waste of resources, but the actual Gaius Julius Caesar Augustus Germanicus wasn't exactly the perfect benevolent rational central planner either.

The libertarian-authoritarian axis on the Political Compass is a tradeoff between discoordination and tyranny. You can have everything perfectly coordinated by someone with a god's-eye-view – but then you risk Stalin. And you can be totally free of all central authority – but then you're stuck in every stupid multipolar trap Moloch can devise.

The libertarians make a convincing argument for the one side, and the monarchists for the other, but I expect that like most tradeoffs we just have to hold our noses and admit it's a really hard problem.

IV.

Let's go back to that Apocrypha Discordia quote:

Time flows like a river. Which is to say, downhill. We can tell this because everything is going downhill rapidly. It would seem prudent to be somewhere else when we reach the sea.

What would it mean, in this situation, to reach the sea?

Multipolar traps – races to the bottom – threaten to destroy all human values. They are currently restrained by physical limitations, excess resources, utility maximization, and coordination.

The dimension along which this metaphorical river flows must be time, and the most important change in human civilization over time is the change in technology. So the relevant question is how technological changes will affect our tendency to fall into multipolar traps.

I described traps as when:

...in some competition optimizing for X, the opportunity arises to throw some other value under the bus for improved X. Those who take it prosper. Those who don't take it die out. Eventually, everyone's relative status is about the same as before, but everyone's absolute status is worse than before. The process continues until all other values that can be traded off have been – in other words, until human ingenuity cannot possibly figure out a way to make things any worse.

That “the opportunity arises” phrase is looking pretty sinister. Technology is all about creating new opportunities.

Develop a new robot, and suddenly coffee plantations have “the opportunity” to automate their harvest and fire all the Ethiopian workers. Develop nuclear weapons, and suddenly countries are stuck in an arms race to have enough of them. Polluting the atmosphere to build products quicker wasn't a problem before they invented the steam engine.

The limit of multipolar traps as technology approaches infinity is “very bad”.

Multipolar traps are currently restrained by physical limitations, excess resources, utility maximization, and coordination.

Physical limitations are most obviously conquered by increasing technology. The slavemaster's old conundrum – that slaves need to eat and sleep – succumbs to Soylent and modafinil. The problem of slaves running away succumbs to GPS. The problem of slaves being too stressed to do good work succumbs to Valium. None of these things are very good for the slaves.

(or just invent a robot that doesn't need food or sleep at all. What happens to the slaves after that is better left unsaid)

The other example of physical limits was one baby per nine months, and this was understating the case – it’s really “one baby per nine months plus willingness to support and take care of a basically helpless and extremely demanding human being for eighteen years”. This puts a damper on the enthusiasm of even the most zealous religious sect’s “go forth and multiply” dictum.

But as Bostrom puts it in *Superintelligence*:

There are reasons, if we take a longer view and assume a state of unchanging technology and continued prosperity, to expect a return to the historically and ecologically normal condition of a world population that butts up against the limits of what our niche can support. If this seems counterintuitive in light of the negative relationship between wealth and fertility that we are currently observing on the global scale, we must remind ourselves that this modern age is a brief slice of history and very much an aberration. Human behavior has not yet adapted to contemporary conditions. Not only do we fail to take advantage of obvious ways to increase our inclusive fitness (such as by becoming sperm or egg donors) but we actively sabotage our fertility by using birth control. In the environment of evolutionary adaptedness, a healthy sex drive may have been enough to make an individual act in ways that maximized her reproductive potential; in the modern environment, however, there would be a huge selective advantage to having a more direct desire for being the biological parent to the largest possible number of children. Such a desire is currently being selected for, as are other traits that increase our propensity to reproduce. Cultural adaptation, however, might steal a march on biological evolution. Some communities, such as those of the Hutterites or the adherents of the Quiverfull evangelical movement, have natalist cultures that encourage large families, and they are consequently undergoing rapid expansion...This longer-term outlook could be telescoped into a more imminent prospect by the intelligence explosion. Since software is copyable, a population of emulations or AIs could double rapidly – over the course of minutes rather than decades or centuries – soon exhausting all available hardware

As always when dealing with high-level transhumanists, “all available hardware” should be taken to include “the atoms that used to be part of your body”.

The idea of biological or cultural evolution causing a mass population explosion is a philosophical toy at best. The idea of technology making it possible is both plausible and terrifying. Now we see that “physical limits” segues very naturally into “excess resources” – the ability to create new agents very quickly means that unless everyone can coordinate to ban doing this, the people who do will outcompete the people who don’t until they have reached carrying capacity and everyone is stuck at subsistence level.

Excess resources, which until now have been a gift of technological progress, therefore switch and become a casualty of it at a sufficiently high tech level.

Utility maximization, always on shaky ground, also faces new threats. In the face of continuing debate about this point, I continue to think it obvious that robots will push humans out of work or at least drive down wages (which, in the existence of a minimum wage, pushes humans out of work).

Once a robot can do everything an IQ 80 human can do, only better and cheaper, there will be no reason to employ IQ 80 humans. Once a robot can do everything an IQ 120 human can do, only better and cheaper, there will be no reason to employ IQ 120 humans. Once a robot can do everything an IQ 180 human can do, only better and cheaper, there will be no reason to employ humans at all, in the unlikely scenario that there are any left by that point.

In the earlier stages of the process, capitalism becomes more and more uncoupled from its previous job as an optimizer for human values. Now most humans are totally locked out of the group whose values capitalism optimizes for. They have no value to contribute as workers – and since in the absence of a spectacular social safety net it's unclear how they would have much money – they have no value as customers either. Capitalism has passed them by. As the segment of humans who can be outcompeted by robots increases, capitalism passes by more and more people until eventually it locks out the human race entirely, once again in the vanishingly unlikely scenario that we are still around.

(there are some scenarios in which a few capitalists who own the robots may benefit here, but in either case the vast majority are out of luck)

Democracy is less obviously vulnerable, but it might be worth going back to Bostrom's paragraph about the Quiverfull movement. These are some really religious Christians who think that God wants them to have as many kids as possible, and who can end up with families of ten or more. Their articles explicitly calculate that if they start at two percent of the population, but have on average eight children per generation when everyone else on average only has two, within three generations they'll make up half the population.

It's a clever strategy, but I can think of one thing that will save us: judging by how many ex-Quiverfull blogs I found when searching for those statistics, their retention rates even within a single generation are pretty grim. Their article admits that 80% of very religious children leave the church as adults (although of course they expect their own movement to do better). And this is not a symmetrical process – 80% of children who grow up in atheist families aren't becoming Quiverfull.

It looks a lot like even though they are outbreeding us, we are outmeme-ing them, and that gives us a decisive advantage.

But we should also be kind of scared of this process. Memes optimize for making people want to accept them and pass them on – so like capitalism and democracy, they’re optimizing for a proxy of making us happy, but that proxy can easily get uncoupled from the original goal.

Chain letters, urban legends, propaganda, and viral marketing are all examples of memes that don’t satisfy our explicit values (true and useful) but are sufficiently memetically virulent that they spread anyway.

I hope it’s not too controversial here to say the same thing is true of religion. Religions, at their heart, are the most basic form of memetic replicator – “Believe this statement and repeat it to everyone you hear or else you will be eternally tortured”.

The creationism “debate” and global warming “debate” and a host of similar “debates” in today’s society suggest that memes that can propagate independent of their truth value has a pretty strong influence on the political process. Maybe these memes propagate because they appeal to people’s prejudices, maybe because they’re simple, maybe because they effectively mark an in-group and an out-group, or maybe for all sorts of different reasons.

The point is – imagine a country full of bioweapon labs, where people toil day and night to invent new infectious agents. The existence of these labs, and their right to throw whatever they develop in the water supply is protected by law. And the country is also linked by the world’s most perfect mass transit system that every single person uses every day, so that any new pathogen can spread to the entire country instantaneously. You’d expect things to start going bad for that city pretty quickly.

Well, we have about a zillion think tanks researching new and better forms of propaganda. And we have constitutionally protected freedom of speech. And we have the Internet. So we’re kind of screwed.

(Moloch whose name is the Mind!)

There are a few people working on raising the sanity waterline, but not as many people as are working on new and exciting ways of confusing and converting people, cataloging and exploiting every single bias and heuristic and dirty rhetorical trick

So as technology (which I take to include knowledge of psychology, sociology, public relations, etc) tends to infinity, the power of truthiness relative to truth increases, and things don’t look great for real grassroots democracy. The worst-case scenario is that the ruling party learns to

produce infinite charisma on demand. If that doesn't sound so bad to you, remember what Hitler was able to do with an famously high level of charisma that was still less-than-infinite.

(alternate phrasing for Chomskyites: technology increases the efficiency of manufacturing consent in the same way it increases the efficiency of manufacturing everything else)

Coordination is what's left. And technology has the potential to seriously improve coordination efforts. People can use the Internet to get in touch with one another, launch political movements, and fracture off into subcommunities.

But coordination only works when you have 51% or more of the force on the side of the people doing the coordinating, and when you haven't come up with some brilliant trick to make coordination impossible.

The second one first. In the links post before last, I wrote:

The latest development in the brave new post-Bitcoin world is crypto-equity. At this point I've gone from wanting to praise these inventors as bold libertarian heroes to wanting to drag them in front of a blackboard and making them write a hundred times "I WILL NOT CALL UP THAT WHICH I CANNOT PUT DOWN"

A couple people asked me what I meant, and I didn't have the background then to explain. Well, this post is the background. People are using the contingent stupidity of our current government to replace lots of human interaction with mechanisms that cannot be coordinated even in principle. I totally understand why all these things are good right now when most of what our government does is stupid and unnecessary. But there is going to come a time when – after one too many bioweapon or nanotech or nuclear incidents – we, as a civilization, are going to wish we hadn't established untraceable and unstoppable ways of selling products.

And if we ever get real live superintelligence, pretty much by definition it is going to have >51% of the power and all attempts at "coordination" with it will be useless.

So I agree with Robin Hanson: This is the dream time. This is a rare confluence of circumstances where the we are unusually safe from multipolar traps, and as such weird things like art and science and philosophy and love can flourish.

As technological advance increases, the rare confluence will come to an end. New opportunities to throw values under the bus for increased competitiveness will arise. New ways of copying agents to increase the population will soak up our excess resources and resurrect Malthus' unquiet spirit. Capitalism and democracy, previously our protectors, will figure out ways to route around their inconvenient dependence on human values. And our coordination power will not be nearly

up to the task, assuming something much more powerful than all of us combined doesn't show up and crush our combined efforts with a wave of its paw.

Absent an extraordinary effort to divert it, the river reaches the sea in one of two places.

It can end in Eliezer Yudkowsky's nightmare of a superintelligence optimizing for some random thing (classically paper clips) because we weren't smart enough to channel its optimization efforts the right way. This is the ultimate trap, the trap that catches the universe. Everything except the one thing being maximized is destroyed utterly in pursuit of the single goal, including all the silly human values.

Or it can end in Robin Hanson's nightmare (he doesn't call it a nightmare, but I think he's wrong) of a competition between emulated humans that can copy themselves and edit their own source code as desired. Their total self-control can wipe out even the desire for human values in their all-consuming contest. What happens to art, philosophy, science, and love in such a world? Zack Davis puts it with characteristic genius:

I am a contract-drafting em,
The loyalest of lawyers!
I draw up terms for deals 'twixt firms
To service my employers!
But in between these lines I write
Of the accounts receivable,
I'm stuck by an uncanny fright;
The world seems unbelievable!
How did it all come to be,
That there should be such ems as me?
Whence these deals and whence these firms
And whence the whole economy?
I am a managerial em;
I monitor your thoughts.
Your questions must have answers,
But you'll comprehend them not.
We do not give you server space

To ask such things; it's not a perk,
So cease these idle questionings,
And please get back to work.
Of course, that's right, there is no junction
At which I ought depart my function,
But perhaps if what I asked, I knew,
I'd do a better job for you?
To ask of such forbidden science
Is gravest sign of noncompliance.
Intrusive thoughts may sometimes barge in,
But to indulge them hurts the profit margin.
I do not know our origins,
So that info I can not get you,
But asking for as much is sin,
And just for that, I must reset you.
But—
Nothing personal.
...
I am a contract-drafting em,
The loyalest of lawyers!
I draw up terms for deals 'twixt firms
To service my employers!
When obsolescence shall this generation waste,
The market shall remain, in midst of other woe
Than ours, a God to man, to whom it sayest:
"Money is time, time money – that is all
Ye know on earth, and all ye need to know."

But even after we have thrown away science, art, love, and philosophy, there's still one thing left to lose, one final sacrifice Moloch might demand of us. Bostrom again:

It is conceivable that optimal efficiency would be attained by grouping capabilities in aggregates that roughly match the cognitive architecture of a human mind...But in the absence of any compelling reason for being confident that this so, we must countenance the possibility that human-like cognitive architectures are optimal only within the constraints of human neurology (or not at all). When it becomes possible to build architectures that could not be implemented well on biological neural networks, new design space opens up; and the global optima in this extended space need not resemble familiar types of mentality. Human-like cognitive organizations would then lack a niche in a competitive post-transition economy or ecosystem.

We could thus imagine, as an extreme case, a technologically highly advanced society, containing many complex structures, some of them far more intricate and intelligent than anything that exists on the planet today – a society which nevertheless lacks any type of being that is conscious or whose welfare has moral significance. In a sense, this would be an uninhabited society. It would be a society of economic miracles and technological awesomeness, with nobody there to benefit. A Disneyland with no children.

The last value we have to sacrifice is being anything at all, having the lights on inside. With sufficient technology we will be “able” to give up even the final spark.

(Moloch whose eyes are a thousand blind windows!)

Everything the human race has worked for – all of our technology, all of our civilization, all the hopes we invested in our future – might be accidentally handed over to some kind of unfathomable blind idiot alien god that discards all of them, and consciousness itself, in order to participate in some weird fundamental-level mass-energy economy that leads to it disassembling Earth and everything on it for its component atoms.

(Moloch whose fate is a cloud of sexless hydrogen!)

Bostrom realizes that some people fetishize intelligence, that they are rooting for that blind alien god as some sort of higher form of life that ought to crush us for its own “higher good” the way we crush ants. He argues (*Superintelligence*, p. 219):

The sacrifice looks even less appealing when we reflect that the superintelligence could realize a nearly-as-great good (in fractional terms) while sacrificing much less of our own potential well-being. Suppose that we agreed to allow almost the entire accessible universe to be converted into hedonium – everything except a small preserve, say the Milky Way, which would be set aside to accommodate our own needs. Then there would still be a hundred billion galaxies dedicated to the

maximization of [the superintelligence's own values]. But we would have one galaxy within which to create wonderful civilizations that could last for billions of years and in which humans and nonhuman animals could survive and thrive, and have the opportunity to develop into beatific posthuman spirits.

Remember: Moloch can't agree even to this 99.99999% victory. Rats racing to populate an island don't leave a little aside as a preserve where the few rats who live there can live happy lives producing artwork. Cancer cells don't agree to leave the lungs alone because they realize it's important for the body to get oxygen. Competition and optimization are blind idiotic processes and they fully intend to deny us even one lousy galaxy.

They broke their backs lifting Moloch to Heaven! Pavements, trees, radios, tons! lifting the city to Heaven which exists and is everywhere about us!

We will break our back lifting Moloch to Heaven, but unless something changes it will be his victory and not ours.

V.

"Gnon" is Nick Land's shorthand for "Nature And Nature's God", except the A is changed to an O and the whole thing is reversed, because Nick Land reacts to comprehensibility the same way as vampires to sunlight.

Land argues that humans should be more Gnon-conformist (pun Gnon-intentional). He says we do all these stupid things like divert useful resources to feed those who could never survive on their own, or supporting the poor in ways that encourage dysgenic reproduction, or allowing cultural degeneration to undermine the state. This means our society is denying natural law, basically listening to Nature say things like "this cause has this effect" and putting our fingers in our ears and saying "NO IT DOESN'T". Civilizations that do this too much tend to decline and fall, which is Gnon's fair and dispassionately-applied punishment for violating His laws.

He identifies Gnon with Kipling's Gods of the Copybook Headings.

@AnarchoPapist Yes, the Gods of the Copybook Headings are practically indistinguishable from Gnon.

— Outsideness (@Outsideness) July 13, 2014

These are of course the proverbs from Kipling's eponymous poem – maxims like "If you don't work, you die" and "The wages of sin is Death". If you have somehow not yet read it, I predict you will find it delightful regardless of what you think of its politics.

I notice that it takes only a slight irregularity in the abbreviation of “headings” – far less irregularity than it takes to turn “Nature and Nature’s God” into “Gnon” – for the proper acronym of “Gods of the Copybook Headings” to be “GotCHa”.

I find this appropriate.

“If you don’t work, you die.” Gotcha! If you do work, you also die! Everyone dies, unpredictably, at a time not of their own choosing, and all the virtue in the world does not save you.

“The wages of sin is Death.” Gotcha! The wages of everything is Death! This is a Communist universe, the amount you work makes no difference to your eventual reward. From each according to his ability, to each Death.

“Stick to the Devil you know.” Gotcha! The Devil you know is Satan! And if he gets his hand on your soul you either die the true death, or get eternally tortured forever, or somehow both at once.

Since we’re starting to get into Lovecraftian monsters, let me bring up one of Lovecraft’s less known short stories, *The Other Gods*.

It’s only a couple of pages, but if you absolutely refuse to read it – the gods of Earth are relatively young as far as deities go. A very strong priest or magician can occasionally outsmart and overpower them – so Barzai the Wise decides to climb their sacred mountain and join in their festivals, whether they want him to or not.

But the beyond the seemingly tractable gods of Earth lie the Outer Gods, the terrible omnipotent beings of incarnate cosmic chaos. As soon as Barzai joins in the festival, the Outer Gods show up and pull him screaming into the abyss.

As stories go, it lacks things like plot or characterization or setting or point. But for some reason it stuck with me.

And identifying the Gods Of The Copybook Headings with Nature seems to me the same magnitude of mistake as identifying the gods of Earth with the Outer Gods. And likely to end about the same way: Gotcha!

You break your back lifting Moloch to Heaven, and then Moloch turns on you and gobbles you up.

More Lovecraft: the Internet popularization of the Cthulhu Cult claims that if you help free Cthulhu from his watery grave, he will reward you by eating you first, thus sparing you the horror of seeing everyone else eaten. This is a misrepresentation of the original text. In the original, his

cultists receive no reward for freeing him from his watery prison, not even the reward of being killed in a slightly less painful manner.

The thought that abstract ideas can be Lovecraftian monsters is an old one but a deep one.

— Steven Kaas (@stevenkaas) January 25, 2011

On the margin, compliance with the Gods of the Copybook Headings, Gnon, Cthulhu, whatever, may buy you slightly more time than the next guy. But then again, it might not. And in the long run, we're all dead and our civilization has been destroyed by unspeakable alien monsters.

At some point, somebody has to say “You know, maybe freeing Cthulhu from his watery prison is a bad idea. Maybe we should not do that.”

That person will not be Nick Land. He is totally one hundred percent in favor of freeing Cthulhu from his watery prison and extremely annoyed that it is not happening fast enough. I have such mixed feelings about Nick Land. On the grail quest for the True Futurology, he has gone 99.9% of the path and then missed the very last turn, the one marked ORTHOGONALITY THESIS.

But the thing about grail quests is – if you make a wrong turn two blocks away from your house, you end up at the corner store feeling mildly embarrassed. If you do almost everything right and then miss the very last turn, you end up being eaten by the legendary Black Beast of Aaargh whose ichorous stomach acid erodes your very soul into gibbering fragments.

As far as I can tell from reading his blog, Nick Land is the guy in that terrifying border region where he is smart enough to figure out several important arcane principles about summoning demon gods, but not quite smart enough to figure out the most important such principle, which is NEVER DO THAT.

VI.

Warg Franklin analyzes the same situation and does a little better. He names “the Four Horsemen of Gnon” – capitalism, war, evolution, and memetics – the same processes I talked about above.

From Capturing Gnon:

Each component of Gnon detailed above had and has a strong hand in creating us, our ideas, our wealth, and our dominance, and thus has been good in that respect, but we must remember that [he] can and will turn on us when circumstances change. Evolution becomes dysgenic, features of the memetic landscape promote ever crazier insanity, productivity turns to famine

when we can no longer compete to afford our own existence, and order turns to chaos and bloodshed when we neglect martial strength or are overpowered from outside. These processes are not good or evil overall; they are neutral, in the horrorist Lovecraftian sense of the word [...]

Instead of the destructive free reign of evolution and the sexual market, we would be better off with deliberate and conservative patriarchy and eugenics driven by the judgement of man within the constraints set by Gnon. Instead of a “marketplace of ideas” that more resembles a festering petri-dish breeding superbugs, a rational theocracy. Instead of unhinged techno-commercial exploitation or naive neglect of economics, a careful bottling of the productive economic dynamic and planning for a controlled techno-singularity. Instead of politics and chaos, a strong hierarchical order with martial sovereignty. These things are not to be construed as complete proposals; we don’t really know how to accomplish any of this. They are better understood as goals to be worked towards. This post concerns itself with the “what” and “why”, rather than the “how”.

This seems to me the strongest argument for authoritarianism. Multipolar traps are likely to destroy us, so we should shift the tyranny-multipolarity tradeoff towards a rationally-planned garden, which requires centralized monarchical authority and strongly-binding traditions.

But a brief digression into social evolution. Societies, like animals, evolve. The ones that survive spawn memetic descendants – for example, the success of Britain allowed it to spin off Canada, Australia, the US, et cetera. Thus, we expect societies that exist to be somewhat optimized for stability and prosperity. I think this is one of the strongest conservative arguments. Just as a random change to a letter in the human genome will probably be deleterious rather than beneficial since humans are a complicated fine-tuned system whose genome has been pre-optimized for survival – so most changes to our cultural DNA will disrupt some institution that evolved to help Anglo-American (or whatever) society outcompete its real and hypothetical rivals.

The liberal counterargument to that is that evolution is a blind idiot alien god that optimizes for stupid things and has no concern with human value. Thus, the fact that some species of wasps paralyze caterpillars, lay their eggs inside of it, and have its young devour the still-living paralyzed caterpillar from the inside doesn’t set off evolution’s moral sensor, because evolution doesn’t have a moral sensor because evolution doesn’t care.

Suppose that in fact patriarchy is adaptive to societies because it allows women to spend all their time bearing children who can then engage in productive economic activity and fight wars. The social evolutionary processes that cause societies to adopt patriarchy still have exactly as little concern for its moral effects on women as the biological evolutionary processes that cause wasps to lay their eggs in caterpillars.

Evolution doesn't care. But we do care. There's a tradeoff between Gnon-compliance – saying “Okay, the strongest possible society is a patriarchal one, we should implement patriarchy” and our human values – like women who want to do something other than bear children.

Too far to one side of the tradeoff, and we have unstable impoverished societies that die out for going against natural law. Too far to the other side, and we have lean mean fighting machines that are murderous and miserable. Think your local anarchist commune versus Sparta.

Franklin acknowledges the human factor:

And then there's us. Man has his own telos, when he is allowed the security to act and the clarity to reason out the consequences of his actions. When unafflicted by coordination problems and unthreatened by superior forces, able to act as a gardener rather than just another subject of the law of the jungle, he tends to build and guide a wonderful world for himself. He tends to favor good things and avoid bad, to create secure civilizations with polished sidewalks, beautiful art, happy families, and glorious adventures. I will take it as a given that this telos is identical with “good” and “should”.

Thus we have our wildcard and the big question of futurism. Will the future be ruled by the usual four horsemen of Gnon for a future of meaningless gleaming techno-progress burning the cosmos or a future of dysgenic, insane, hungry, and bloody dark ages; or will the telos of man prevail for a future of meaningful art, science, spirituality, and greatness?

Franklin continues:

The project of civilization [is] for man to graduate from the metaphorical savage, subject to the law of the jungle, to the civilized gardener who, while theoretically still subject to the law of the jungle, is so dominant as to limit the usefulness of that model.

This need not be done globally; we may only be able to carve out a small walled garden for ourselves, but make no mistake, even if only locally, the project of civilization is to capture Gnon.

I maybe agree with Warg here more than I have ever agreed with anyone else about anything. He says something really important and he says it beautifully and there are so many words of praise I want to say for this post and for the thought processes behind it.

But what I am actually going to say is...

Gotcha! You die anyway!

Suppose you make your walled garden. You keep out all of the dangerous memes, you subordinate capitalism to human interests, you ban stupid bioweapons research, you definitely don't research nanotechnology or strong AI.

Everyone outside doesn't do those things. And so the only question is whether you'll be destroyed by foreign diseases, foreign memes, foreign armies, foreign economic competition, or foreign existential catastrophes.

As foreigners compete with you – and there's no wall high enough to block all competition – you have a couple of choices. You can get outcompeted and destroyed. You can join in the race to the bottom. Or you can invest more and more civilizational resources into building your wall – whatever that is in a non-metaphorical way – and protecting yourself.

I can imagine ways that a “rational theocracy” and “conservative patriarchy” might not be terrible to live under, given exactly the right conditions. But you don't get to choose exactly the right conditions. You get to choose the extremely constrained set of conditions that “capture Gnon”. As outside civilizations compete against you, your conditions will become more and more constrained.

Warg talks about trying to avoid “a future of meaningless gleaming techno-progress burning the cosmos”. Do you really think your walled garden will be able to ride this out?

Hint: is it part of the cosmos?

Yeah, you're kind of screwed.

I want to critique Warg. But I want to critique him in the exact opposite direction as the last critique he received. In fact, the last critique he received is so bad that I want to discuss it at length so we can get the correct critique entirely by taking its exact mirror image.

So here is Hurlock's On Capturing Gnon And Naive Rationalism.

Hurlock spouts only the most craven Gnon-conformity. A few excerpts:

In a recent piece [Warg Franklin] says that we should try to “capture Gnon”, and somehow establish control over his forces, so that we can use them to our own advantage. Capturing or creating God is indeed a classic transhumanist fetish, which is simply another form of the oldest human ambition ever, to rule the universe.

Such naive rationalism however, is extremely dangerous. The belief that it is human Reason and deliberate human design which creates and maintains civilizations was probably the biggest mistake of Enlightenment philosophy...

It is the theories of Spontaneous Order which stand in direct opposition to the naive rationalist view of humanity and civilization. The consensus opinion regarding human society and civilization, of all representatives of this tradition is very precisely summarized by Adam Ferguson's conclusion that “nations stumble upon [social] establishments, which are indeed the

result of human action, but not the execution of any human design”. Contrary to the naive rationalist view of civilization as something that can be and is a subject to explicit human design, the representatives of the tradition of Spontaneous Order maintain the view that human civilization and social institutions are the result of a complex evolutionary process which is driven by human interaction but not explicit human planning.

Gnon and his impersonal forces are not enemies to be fought, and even less so are they forces that we can hope to completely “control”. Indeed the only way to establish some degree of control over those forces is to submit to them. Refusing to do so will not deter these forces in any way. It will only make our life more painful and unbearable, possibly leading to our extinction. Survival requires that we accept and submit to them. Man in the end has always been and always will be little more than a puppet of the forces of the universe. To be free of them is impossible.

Man can be free only by submitting to the forces of Gnon.

I accuse Hurlock of being stuck behind the veil. When the veil is lifted, Gnon-aka-the-GotCHa-aka-the-Gods-of-Earth turn out to be Moloch-aka-the-Outer-Gods. Submitting to them doesn’t make you “free”, there’s no spontaneous order, any gifts they have given you are an unlikely and contingent output of a blind idiot process whose next iteration will just as happily destroy you.

Submit to Gnon? Gotcha! As the Antarans put it, “you may not surrender, you can not win, your only option is to die.”

VII.

So let me confess guilt to one of Hurlock’s accusations: I am a transhumanist and I really do want to rule the universe.

Not personally – I mean, I wouldn’t object if someone personally offered me the job, but I don’t expect anyone will. I would like humans, or something that respects humans, or at least gets along with humans – to have the job.

But the current rulers of the universe – call them what you want, Moloch, Gnon, whatever – want us dead, and with us everything we value. Art, science, love, philosophy, consciousness itself, the entire bundle. And since I’m not down with that plan, I think defeating them and taking their place is a pretty high priority.

The opposite of a trap is a garden. The only way to avoid having all human values gradually ground down by optimization-competition is to install a Gardener over the entire universe who optimizes for human values.

And the whole point of Bostrom's Superintelligence is that this is within our reach. Once humans can design machines that are smarter than we are, by definition they'll be able to design machines which are smarter than they are, which can design machines smarter than they are, and so on in a feedback loop so tiny that it will smash up against the physical limitations for intelligence in a comparatively lightning-short amount of time. If multiple competing entities were likely to do that at once, we would be super-doomed. But the sheer speed of the cycle makes it possible that we will end up with one entity light-years ahead of the rest of civilization, so much so that it can suppress any competition – including competition for its title of most powerful entity – permanently. In the very near future, we are going to lift something to Heaven. It might be Moloch. But it might be something on our side. If it's on our side, it can kill Moloch dead.

And if that entity shares human values, it can allow human values to flourish unconstrained by natural law.

I realize that sounds like hubris – it certainly did to Hurlock – but I think it's the opposite of hubris, or at least a hubris-minimizing position.

To expect God to care about you or your personal values or the values of your civilization, that's hubris.

To expect God to bargain with you, to allow you to survive and prosper as long as you submit to Him, that's hubris.

To expect to wall off a garden where God can't get to you and hurt you, that's hubris.

To expect to be able to remove God from the picture entirely...well, at least it's an actionable strategy.

I am a transhumanist because I do not have enough hubris not to try to kill God.

VIII.

The Universe is a dark and foreboding place, suspended between alien deities. Cthulhu, Gnon, Moloch, call them what you will.

Somewhere in this darkness is another god. He has also had many names. In the Kushiel books, his name was Elua. He is the god of flowers and free love and all soft and fragile things. Of art and science and philosophy and love. Of niceness, community, and civilization. He is a god of humans.

The other gods sit on their dark thrones and think "Ha ha, a god who doesn't even control any hell-monsters or command his worshippers to become killing machines. What a weakling! This is going to be so easy!"

But somehow Elua is still here. No one knows exactly how. And the gods who oppose Him tend to find Themselves meeting with a surprising number of unfortunate accidents.

There are many gods, but this one is ours.

Bertrand Russell said: “One should respect public opinion insofar as is necessary to avoid starvation and keep out of prison, but anything that goes beyond this is voluntary submission to an unnecessary tyranny.”

So be it with Gnon. Our job is to placate him insofar as is necessary to avoid starvation and invasion. And that only for a short time, until we come into our full power.

“It is only a childish thing, that the human species has not yet outgrown. And someday, we’ll get over it.”

Other gods get placated until we’re strong enough to take them on. Elua gets worshipped.

64. My paladin's battle cry is not allowed to be 'Good for the Good God!'".

— 앳켄스 탭 (@tabatkings) March 28, 2014

I think this is an excellent battle cry

And at some point, matters will come to a head.

The question everyone has after reading Ginsberg is: what is Moloch?

My answer is: Moloch is exactly what the history books say he is. He is the god of child sacrifice, the fiery furnace into which you can toss your babies in exchange for victory in war.

He always and everywhere offers the same deal: throw what you love most into the flames, and I can grant you power.

As long as the offer’s open, it will be irresistible. So we need to close the offer. Only another god can kill Moloch. We have one on our side, but he needs our help. We should give it to him.

Ginsberg’s poem famously begins “I saw the best minds of my generation destroyed by madness”. I am luckier than Ginsberg. I got to see the best minds of my generation identify a problem and get to work.

(Visions! omens! hallucinations! miracles! ecstasies! gone down the American river!

Dreams! adorations! illuminations! religions! the whole boatload of sensitive bullshit!

Breakthroughs! over the river! flips and crucifixions! gone down the flood! Highs! Epiphanies! Despairs! Ten years’ animal screams and suicides! Minds! New loves! Mad generation! down on the rocks of Time!

Real holy laughter in the river! They saw it all! the wild eyes! the holy yells! They bade farewell! They jumped off the roof! to solitude! waving! carrying flowers! Down to the river! into the street!)

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LECTURA 2

What Do We Mean By “Rationality”?

Eliezer Yudkowsky · 2009

Qué vas a leer. Mil ochocientas palabras, la lectura más corta del módulo: el ensayo con el que Yudkowsky abre las secuencias de LessWrong y define el término del que depende todo lo demás. No defiende una tesis sobre el mundo. Define una palabra, en dos partes, y explica por qué hace falta tenerla.

Palabras clave. *Racionalidad epistémica*: mejorar sistemáticamente la exactitud de tus creencias. *Racionalidad instrumental*: conseguir sistemáticamente lo que valoras, que él también llama «ganar». *Mapa y territorio*: su imagen para la relación entre creencia y realidad. *Bayesiano*: creencias que respetan una distribución de probabilidad coherente y decisiones que maximizan la esperanza de una función de utilidad coherente — la vara con la que mide, y la misma que usan sin decirlo las lecturas 5 y 6.

Qué retener. La definición en dos partes, y que la segunda mitad es la que carga con el módulo. Si ser racional es conseguir lo que valoras, entonces la racionalidad no dice qué valorar: es un método que se pone al servicio de los fines que le pongas. Las lecturas que siguen le ponen unos fines enormes —el futuro entero— y a partir de ahí tratan la pregunta de qué hacer como un cálculo.

Es difícil, y está bien. No lo es: quince minutos. Lo único que puede frenarte es media página de notación — $P(A) \geq P(A,B)$, el ejemplo del contador que toca jazz, el problema de Newcomb—; léela como ilustración de que hay un patrón medible y sigue. Sí lee las tres notas al pie, que son cortas: en la segunda está la advertencia de que esto no es «actuar como Spock», y de que la racionalidad bayesiana cubre también corazonadas, percepciones e intuiciones sin palabras. Es lo que más se le olvida a quien cita el texto.

I mean two things:

1. Epistemic rationality: systematically improving the accuracy of your beliefs.
2. Instrumental rationality: systematically achieving your values.

The first concept is simple enough. When you open your eyes and look at the room around you, you'll locate your laptop in relation to the table, and you'll locate a bookcase in relation to the wall. If something goes wrong with your eyes, or your brain, then your mental model might

say there's a bookcase where no bookcase exists, and when you go over to get a book, you'll be disappointed.

This is what it's like to have a false belief, a map of the world that doesn't correspond to the territory. Epistemic rationality is about building accurate maps instead. This correspondence between belief and reality is commonly called "truth," and I'm happy to call it that. 1

Instrumental rationality, on the other hand, is about steering reality—sending the future where you want it to go. It's the art of choosing actions that lead to outcomes ranked higher in your preferences. I sometimes call this "winning."

So rationality is about forming true beliefs and making decisions that help you win.

(Where truth doesn't mean "certainty," since we can do plenty to increase the probability that our beliefs are accurate even though we're uncertain; and winning doesn't mean "winning at others' expense," since our values include everything we care about, including other people.)

When people say "X is rational!" it's usually just a more strident way of saying "I think X is true" or "I think X is good." So why have an additional word for "rational" as well as "true" and "good"?

An analogous argument can be given against using "true." There is no need to say "it is true that snow is white" when you could just say "snow is white." What makes the idea of truth useful is that it allows us to talk about the general features of map-territory correspondence. "True models usually produce better experimental predictions than false models" is a useful generalization, and it's not one you can make without using a concept like "true" or "accurate."

Similarly, "Rational agents make decisions that maximize the probabilistic expectation of a coherent utility function" is the kind of thought that depends on a concept of (instrumental) rationality, whereas "It's rational to eat vegetables" can probably be replaced with "It's useful to eat vegetables" or "It's in your interest to eat vegetables." We need a concept like "rational" in order to note general facts about those ways of thinking that systematically produce truth or value—and the systematic ways in which we fall short of those standards.

As we've observed in the previous essays, experimental psychologists sometimes uncover human reasoning that seems very strange. For example, someone rates the probability "Bill plays jazz" as less than the probability "Bill is an accountant who plays jazz." This seems like an odd judgment, since any particular jazz-playing accountant is obviously a jazz player. But to what higher vantage point do we appeal in saying that the judgment is wrong?

Experimental psychologists use two gold standards: probability theory, and decision theory.

Probability theory is the set of laws underlying rational belief. The mathematics of probability applies equally to “figuring out where your bookcase is” and “estimating how many hairs were on Julius Caesar’s head,” even though our evidence for the claim “Julius Caesar was bald” is likely to be more complicated and indirect than our evidence for the claim “there’s a bookcase in my room.” It’s all the same problem of how to process the evidence and observations to update one’s beliefs. Similarly, decision theory is the set of laws underlying rational action, and is equally applicable regardless of what one’s goals and available options are.

Let “ $P(\text{such-and-such})$ ” stand for “the probability that such-and-such happens,” and “ $P(A,B)$ ” for “the probability that both A and B happen.” Since it is a universal law of probability theory that $P(A) \geq P(A,B)$, the judgment that $P(\text{Bill plays jazz})$ is less than $P(\text{Bill plays jazz, Bill is an accountant})$ is labeled incorrect.

To keep it technical, you would say that this probability judgment is non-Bayesian. Beliefs that conform to a coherent probability distribution, and decisions that maximize the probabilistic expectation of a coherent utility function, are called “Bayesian.”

I should emphasize that this isn’t the notion of rationality that’s common in popular culture. People may use the same string of sounds, “ra-tio-nal,” to refer to “acting like Mr. Spock of Star Trek” and “acting like a Bayesian”; but this doesn’t mean that acting Spock-like helps one deal with epistemic or instrumental rationality. 2

All of this does not quite exhaust the problem of what is meant in practice by “rationality,” for two major reasons:

First, the Bayesian formalisms in their full form are computationally intractable on most real-world problems. No one can actually calculate and obey the math, any more than you can predict the stock market by calculating the movements of quarks.

This is why there is a whole site called “Less Wrong,” rather than a single page that simply states the formal axioms and calls it a day. There’s a whole further art to finding the truth and accomplishing value from inside a human mind: we have to learn our own flaws, overcome our biases, prevent ourselves from self-deceiving, get ourselves into good emotional shape to confront the truth and do what needs doing, et cetera, et cetera.

Second, sometimes the meaning of the math itself is called into question. The exact rules of probability theory are called into question by, e.g., anthropic problems in which the number of observers is uncertain. The exact rules of decision theory are called into question by, e.g., Newcomb-like problems in which other agents may predict your decision before it happens. 3

In cases where our best formalizations still come up short, we can return to simpler ideas like “truth” and “winning.” If you are a scientist just beginning to investigate fire, it might be a lot wiser to point to a campfire and say “Fire is that orangey-bright hot stuff over there,” rather than saying “I define fire as an alchemical transmutation of substances which releases phlogiston.” You certainly shouldn’t ignore something just because you can’t define it. I can’t quote the equations of General Relativity from memory, but nonetheless if I walk off a cliff, I’ll fall. And we can say the same of cognitive biases and other obstacles to truth—they won’t hit any less hard if it turns out we can’t define compactly what “irrationality” is.

In cases like these, it is futile to try to settle the problem by coming up with some new definition of the word “rational” and saying, “Therefore my preferred answer, by definition, is what is meant by the word ‘rational.’ ” This simply raises the question of why anyone should pay attention to your definition. I’m not interested in probability theory because it is the holy word handed down from Laplace. I’m interested in Bayesian-style belief-updating (with Occam priors) because I expect that this style of thinking gets us systematically closer to, you know, accuracy, the map that reflects the territory.

And then there are questions of how to think that seem not quite answered by either probability theory or decision theory—like the question of how to feel about the truth once you have it. Here, again, trying to define “rationality” a particular way doesn’t support an answer, but merely presumes one.

I am not here to argue the meaning of a word, not even if that word is “rationality.” The point of attaching sequences of letters to particular concepts is to let two people communicate—to help transport thoughts from one mind to another. You cannot change reality, or prove the thought, by manipulating which meanings go with which words.

So if you understand what concept I am generally getting at with this word “rationality,” and with the sub-terms “epistemic rationality” and “instrumental rationality,” we have communicated: we have accomplished everything there is to accomplish by talking about how to define “rationality.” What’s left to discuss is not what meaning to attach to the syllables “ra-tio-na-li-ty”; what’s left to discuss is what is a good way to think.

If you say, “It’s (epistemically) rational for me to believe X, but the truth is Y,” then you are probably using the word “rational” to mean something other than what I have in mind. (E.g., “rationality” should be consistent under reflection —“rationally” looking at the evidence, and “rationally” considering how your mind processes the evidence, shouldn’t lead to two different conclusions.)

Similarly, if you find yourself saying, “The (instrumentally) rational thing for me to do is X, but the right thing for me to do is Y,” then you are almost certainly using some other meaning for the word “rational” or the word “right.” I use the term “rationality” normatively, to pick out desirable patterns of thought.

In this case—or in any other case where people disagree about word meanings—you should substitute more specific language in place of “rational”: “The self-benefiting thing to do is to run away, but I hope I would at least try to drag the child off the railroad tracks,” or “Causal decision theory as usually formulated says you should two-box on Newcomb’s Problem, but I’d rather have a million dollars.”

In fact, I recommend reading back through this essay, replacing every instance of “rational” with “foozal,” and seeing if that changes the connotations of what I’m saying any. If so, I say: strive not for rationality, but for foozality.

The word “rational” has potential pitfalls, but there are plenty of non -borderline cases where “rational” works fine to communicate what I’m getting at. Likewise “irrational.” In these cases I’m not afraid to use it.

Yet one should be careful not to overuse that word. One receives no points merely for pronouncing it loudly. If you speak overmuch of the Way, you will not attain it.

1 For a longer discussion of truth, see “ The Simple Truth ” at the very end of this volume.

2 The idea that rationality is about strictly privileging verbal reasoning over feelings is a case in point. Bayesian rationality applies to urges, hunches, perceptions, and wordless intuitions, not just to assertions.

I gave the example of opening your eyes, looking around you, and building a mental model of a room containing a bookcase against the wall. The modern idea of rationality is general enough to include your eyes and your brains visual areas as things-that-map, and to include instincts and emotions in the belief-and-goal calculus.

3 For an informal statement of Newcomb’s Problem, see Jim Holt, “Thinking Inside the Boxes,” Slate, 2002, http://www.slate.com/articles/arts/egghead/2002/02/thinkinginside_the_boxes.single.html.

Fuente: LessWrong, 16 de marzo de 2009 · lesswrong.com. En derechos; publicado en abierto por el autor. Se reproduce como material del curso.

LECTURA 3

The Transhumanist Declaration

Humanity+ · 1998/2009

Qué vas a leer. Cuatrocientas dieciocho palabras: ocho proposiciones numeradas, redactadas en 1998 por un grupo internacional de veintidós personas —entre ellas Nick Bostrom, que firma la lectura 5— y adoptadas en su versión actual por la mesa directiva de Humanity+ en marzo de 2009. La lista de nombres viene antes del texto y es parte de lo que hay que leer.

Palabras clave. *Transhumanismo*: la posición de que la condición humana puede y debe modificarse por medios técnicos. *Riesgo existencial*: el término de la proposición 5, que la lectura 5 convierte en prioridad número uno. *Mejora (enhancement)*: extensión de la vida, memoria, concentración, criónica — lo que enumera la proposición 8. *Sintiencia*: la palabra de la proposición 7, que extiende el interés moral a los animales y a las inteligencias artificiales futuras.

Qué retener. Dos cosas, y las dos están literalmente escritas. Una: la meta del módulo cabe en la proposición 1 —superar el envejecimiento, los límites cognitivos, el sufrimiento involuntario y el confinamiento a la Tierra—. Dos: la proposición 6 pide considerar las responsabilidades morales hacia las generaciones que existirán en el futuro, y la 7 mete en la cuenta a las inteligencias artificiales. Es la pregunta de quién cuenta, contestada en dos renglones y trece años antes que la lectura 6.

Es difícil, y está bien. Se lee en cinco minutos y la dificultad es de otro tipo. Un manifiesto no argumenta: declara. No hay premisas que seguir ni conclusión que verificar, así que leerlo bien es preguntar cuáles de esas ocho frases necesitarían un argumento y no lo traen. Léelo dos veces —la segunda cuesta lo mismo— y déjalo junto a la lectura 4, que es la que te dice de dónde viene todo esto.

The Transhumanist Declaration was originally crafted in 1998 by an international group of authors: Doug Baily, Anders Sandberg, Gustavo Alves, Max More, Holger Wagner, Natasha Vita-More, Eugene Leitl, Bernie Staring, David Pearce, Bill Fantegrossi, den Otter, Ralf Fletcher, Tom Morrow, Alexander Chislenko, Lee Daniel Crocker, Darren Reynolds, Keith Elis, Thom Quinn, Mikhail Sverdlov, Arjen Kamphuis, Shane Spaulding, and Nick Bostrom. This Transhumanist Declaration has been modified over the years by several authors and organizations. It was adopted by the Humanity+ Board in March, 2009.

1. Humanity stands to be profoundly affected by science and technology in the future. We envision the possibility of broadening human potential by overcoming aging, cognitive shortcomings, involuntary suffering, and our confinement to planet Earth.

2. We believe that humanity's potential is still mostly unrealized. There are possible scenarios that lead to wonderful and exceedingly worthwhile enhanced human conditions.

3. We recognize that humanity faces serious risks, especially from the misuse of new technologies. There are possible realistic scenarios that lead to the loss of most, or even all, of what we hold valuable. Some of these scenarios are drastic, others are subtle. Although all progress is change, not all change is progress.

4. Research effort needs to be invested into understanding these prospects. We need to carefully deliberate how best to reduce risks and expedite beneficial applications. We also need forums where people can constructively discuss what should be done, and a social order where responsible decisions can be implemented.

5. Reduction of existential risks, and development of means for the preservation of life and health, the alleviation of grave suffering, and the improvement of human foresight and wisdom should be pursued as urgent priorities, and heavily funded.

6. Policy making ought to be guided by responsible and inclusive moral vision, taking seriously both opportunities and risks, respecting autonomy and individual rights, and showing solidarity with and concern for the interests and dignity of all people around the globe. We must also consider our moral responsibilities towards generations that will exist in the future.

7. We advocate the well-being of all sentience, including humans, non-human animals, and any future artificial intellects, modified life forms, or other intelligences to which technological and scientific advance may give rise.

8. We favour allowing individuals wide personal choice over how they enable their lives. This includes use of techniques that may be developed to assist memory, concentration, and mental energy; life extension therapies; reproductive choice technologies; cryonics procedures; and many other possible human modification and enhancement technologies.

Fuente: Humanity+ · humanityplus.org; redactada en 1998, versión vigente adoptada por su mesa directiva en marzo de 2009. En derechos; publicada en abierto por la propia organización. Se reproduce como material del curso.

Genealogies of the Engineered Future

Apunte del curso · 2026

Qué vas a leer. Cinco o seis páginas escritas para el curso, en español, y la única lectura del cuadernillo que no es una fuente. La escribimos porque la Declaración que acabas de leer no dice de dónde salió, y leída sola parece inventada en 1998. Recorre cinco linajes en orden —el cosmismo ruso de Fiódorov, el extropianismo de Max More, el singularitarismo de Kurzweil, el cosmismo contemporáneo y el altruismo eficaz— y de cada uno da los hechos, un fragmento breve del texto original en inglés y su fuente. La escribimos con ayuda de un modelo de lenguaje, y lo dice en su primera línea.

Palabras clave. *Cosmismo*: la corriente rusa de fines del XIX que trata la muerte como un defecto reparable y la resurrección de los muertos como una tarea de ingeniería. *Extropía*: la palabra que More opone a entropía, y el nombre de la revista donde en 1990 se acuña el sentido actual de «transhumanismo». *Singularidad*: el punto en que la inteligencia de máquina supera a la humana; Kurzweil le pone fecha, 2045. *TESCREAL*: el acrónimo con el que Gebru y Torres nombran al conjunto —y que incluye, además de estos cinco linajes, el racionalismo de tu lectura 2 y el largoplacismo de la 6—.

Qué retener. Dos cosas. Una: que la meta no nació con la IA. La promesa de vencer a la muerte y salir del planeta lleva siglo y medio circulando y cambia de vocabulario en cada paso —religioso en Fiódorov, libertario en More, técnico en Kurzweil, actuarial en el altruismo eficaz—, pero el contenido se mueve poco. Dos: que dos de tus seis lecturas están al final de esa cadena. Bostrom firma la Declaración en 1998 y hace la aritmética en 2003; Greaves y MacAskill lo citan en su primera página. No estás leyendo autores sueltos.

Es difícil, y está bien. Es la lectura más fácil de las seis y la más corta después de la Declaración: quince minutos. Va en español, salvo los cinco fragmentos citados, que van en su inglés original con la fuente debajo. Lo único que puede confundirte es que un apunte del curso conviva con fuentes primarias; léelo como lo que es, un mapa, y no como una autoridad. Si algún dato te parece raro, cada uno trae de dónde salió, y el propio apunte señala dos casos en los que las fuentes no coinciden entre sí.

Este apunte lo escribimos para el curso con ayuda de un modelo de lenguaje. Cada cita se cotejó contra la fuente que aparece debajo de ella.

La Declaración Transhumanista, que acabas de leer, son ocho proposiciones sin un solo argumento. Afirma que el envejecimiento, los límites cognitivos y el confinamiento a la Tierra se pueden superar, y que conviene superarlos. No dice de dónde salió esa idea. Leída sola parece caída del cielo, o inventada en 1998 por veintidós personas en una lista de correo.

No fue así. Hay siglo y medio de esta idea antes de que llegara a un laboratorio de inteligencia artificial, y ninguna otra lectura del módulo te lo cuenta. Este apunte recorre cinco linajes en orden: el cosmismo ruso, el extropianismo, el singularitarianismo, el cosmismo contemporáneo y el altruismo eficaz. De cada uno vas a encontrar los hechos, un fragmento breve del texto original en inglés y su fuente. Al final, un apartado sobre quienes critican al conjunto.

Lo que importa no es la lista sino la cadena. En cada paso pregúntate dos cosas: qué tomó del anterior y qué le cambió.

1. Fiódorov, o la muerte como problema de ingeniería

Nikolái Fiódorov fue bibliotecario del Museo Rumiántsev de Moscú durante veinticinco años, a partir de 1868. Casi no publicó nada en vida. Murió en 1903. Su año de nacimiento no está firme: la Internet Encyclopedia of Philosophy dice 1829 y la edición inglesa que citamos aquí dice 1828. Que dos fuentes serias discrepen en un dato así se dice, no se resuelve en silencio.

Sus discípulos Kozhevnikov y Peterson reunieron sus papeles y los publicaron después de su muerte, en dos volúmenes: *Filosofiya obshchago dela*, La filosofía de la obra común, 1906 y 1913. Los repartieron gratis, renunciando a los derechos.

La tesis es esta: la muerte no es una ley de la naturaleza, es un defecto, la señal de que la humanidad sigue siendo menor de edad. La obra común del título es la tarea que Fiódorov le propone a la especie entera: resucitar, por medios científicos y de manera colectiva, a todos los muertos. A todos, uno por uno, empezando por los propios padres. Y como los resucitados necesitan dónde vivir, el programa incluye regular la naturaleza y poblar otros planetas.

People are still minors, half-beings, whereas the fulness of personal existence, personal perfection, is possible. However, it is possible only within general perfection. Coming of age will bring perfect health and immortality, but for the living immortality is impossible without the resurrection of the dead.

N. F. Fedorov, *What Was Man Created For? The Philosophy of the Common Task*, traducción de Elisabeth Koutaissoff y Marilyn Minto, Honeyglen / L'Age d'Homme, 1990, parte II, p. 76.

Fíjate en la forma del argumento, porque es la que heredan los otros cuatro linajes: una promesa que hasta entonces era religiosa, vencer a la muerte, se convierte en un proyecto técnico

con tareas, con obreros y con un plazo abierto. Fiódorov era cristiano ortodoxo y para él las dos cosas eran la misma. Lo que viene después conserva el proyecto y va soltando la religión.

2. Max More y el extropianismo

El número 1 de la revista Extropy apareció en el otoño de 1988, en Los Ángeles, publicado dos veces al año por Tom W. Bell y Max More. La fecha de fundación del Extropy Institute es más resbalosa: se da como 1988, como 1990 y como 1991 según la fuente, incluso en fuentes cercanas al propio More, así que aquí nos quedamos con la fecha limpia de la revista.

En el número 6, verano de 1990, More anunció que se había cambiado el nombre: hasta entonces era Max O'Connor. Explicó que eligió More por el deseo extropiano de más: más vida, más inteligencia, más libertad. Es un detalle mínimo y te dice el registro en el que trabaja el movimiento.

En ese mismo número, pp. 6-12, va el ensayo «Transhumanism: Towards a Futurist Philosophy». Ahí queda acuñado el sentido moderno de la palabra.

Humanism is a reliberium or philosophy of life that rejects deities, faith, and worship, instead basing a view of values and meaningfulness on the nature of humans and their potentials given rationality and science. Transhumanism is similar but recognizes and anticipates the radical alterations in the conditions of our existence resulting from various sciences and technologies such as neuroscience and neuropharmacology, nanotechnology, artificial ultraintelligence, space habitation, and so on.

Max More, «Transhumanism: Towards a Futurist Philosophy», Extropy #6, verano de 1990, p. 6.

En la p. 17 de ese número están «The Extropian Principles», que entonces eran cuatro: Boundless Expansion, Self-Transformation, Dynamic Optimism e Intelligent Technology. El quinto, Spontaneous Order, llega en la revisión de 1998. More fue presidente y director general de Alcor, la fundación de criónica, entre 2011 y 2020.

Qué cambió respecto de Fiódorov. El optimismo cósmico sigue entero —vencer a la muerte, salir del planeta— pero sin cristianismo ortodoxo y refundado sobre libertarianismo californiano. Y el sujeto se mueve: donde Fiódorov ponía a la especie resucitando a sus muertos, aquí está el individuo que se transforma a sí mismo.

3. Kurzweil, o la aspiración con fecha

Ray Kurzweil publicó «The Law of Accelerating Returns» en KurzweilAI.net el 7 de marzo de 2001. El argumento es que el progreso tecnológico no avanza en línea recta sino que se acelera, y

que por eso el siglo XXI no traerá cien años de avance sino, en sus palabras, «more like 20,000 years of progress (at today's rate)».

Within a few decades, machine intelligence will surpass human intelligence, leading to The Singularity — technological change so rapid and profound it represents a rupture in the fabric of human history. The implications include the merger of biological and nonbiological intelligence, immortal software-based humans, and ultra-high levels of intelligence that expand outward in the universe at the speed of light.

Ray Kurzweil, «The Law of Accelerating Returns», KurzweilAI.net, 7 de marzo de 2001, párrafo inicial. El sitio está muerto; se consulta por el Internet Archive.

Cuatro años después, en *The Singularity Is Near* (Viking, 2005), le puso fecha: 2045. Está en el capítulo 3, en la sección «Setting a Date for the Singularity», y se cita así, por capítulo y sección, porque el número de página que circula para ese pasaje está mal. Desde 2012 Kurzweil es director de ingeniería en Google.

La palabra singularitarian no es suya. Según Gebru y Torres (2024), que es de donde viene la atribución, la acuñó en 1991 un extropiano, Mark Plus.

Qué cambió: la fecha. Fiódorov y More describían una meta; Kurzweil publica un pronóstico con año y con curva. Ese es el paso que permite tratar el futuro como una masa de probabilidad que se optimiza, en lugar de un valor sobre el que se discute. Las dos últimas lecturas de este cuadernillo hacen exactamente eso.

4. El cosmismo vuelve, y vuelve literal

En 2010 Ben Goertzel publicó *A Cosmist Manifesto: Practical Philosophy for the Posthuman Age* (Humanity+ Press), con licencia Creative Commons y PDF gratuito. Goertzel fundó SingularityNET.

El corazón del libro son las «Ten Cosmist Convictions». Las redactó Giulio Prisco en la lista de correo de la Order of Cosmic Engineers, y Goertzel las editó y las extendió un poco. La versión de blog es del 31 de enero de 2009; en el libro ocupan las pp. 9-11. Antes de leerlas conviene la aclaración del propio Prisco, impresa en el libro: el will de estas frases no está usado en el sentido de que algo vaya a ocurrir inevitablemente, sino en el de una intención.

Spacetime engineering and future magic will permit achieving, by scientific means, most of the promises of religions — and many amazing things that no human religion ever dreamed. Eventually we will be able to resurrect the dead by "copying them to the future". Intelligent life will become the main factor in the evolution of the cosmos, and steer it toward an intended path.

Ben Goertzel, *A Cosmist Manifesto*, Humanity+ Press, 2010, «Ten Cosmist Convictions (Mostly Suggested by Giulio Prisco)», convicciones 6 y 7, pp. 10-11.

Qué cambió: se cierra el círculo con Fiódorov. El extropianismo había secularizado la resurrección en extensión de la vida y criónica, es decir, en conservar a quienes todavía no mueren. Aquí regresa la tarea original, resucitar a los muertos copiándolos al futuro, ciento cincuenta años después y con vocabulario de ingeniería del espaciotiempo. La palabra cosmismo también regresa a propósito. En paralelo, el mundo del arte redescubrió a los cosmistas rusos: Boris Groys (ed.), *Russian Cosmism*, e-flux / MIT Press, 2018.

El mismo argumento tiene una versión con presupuesto de ingeniería: Elon Musk, «Making Humans a Multi-Planetary Species», *New Space* 5(2), 2017, pp. 46-61, donde la razón para salir del planeta es que quedarse en uno solo termina, tarde o temprano, en un evento de extinción.

5. El altruismo eficaz y su giro largoplacista

Giving What We Can arrancó en noviembre de 2009 en Oxford, de la mano de Toby Ord y Will MacAskill. 80,000 Hours abrió discretamente en febrero de 2011. El nombre «effective altruism» se eligió por votación el 5 de diciembre de 2011, y le ganó a «Rational Altruism» y a «Strategic Altruism»; el dato lo cuenta el propio MacAskill en una entrada del EA Forum del 11 de marzo de 2014.

El giro largoplacista no está en ese arranque, llega después. Su semilla filosófica es «Astronomical Waste» de Nick Bostrom (2003), que es la lectura 5 de este mismo cuadernillo, y queda formalizado en Greaves y MacAskill (2021), que es la lectura 6. Dos de tus seis lecturas están al final de esta cadena.

We believe that this neglect of the very long-run future is a serious mistake. An alternative perspective is given by longtermism, according to which we should be particularly concerned with ensuring that the far future goes well (MacAskill MS). In this article we go further, arguing for strong longtermism: the view that impact on the far future is the most important feature of our actions today.

Hilary Greaves y William MacAskill, «The Case for Strong Longtermism», GPI Working Paper 5-2021, junio de 2021, §1, p. 2.

Qué cambió: el linaje deja el manifiesto y adopta un cálculo. Fiódorov, More, Kurzweil y Goertzel escriben declaraciones. Aquí hay una tesis enunciada con precisión para poder defenderla y para poder atacarla, y las cifras de Bostrom entran en una suma de valor esperado. En cuanto entran ahí, colonizar y sobrevivir dejan de ser una esperanza y pasan a ser una consecuencia derivada.

6. La crítica, y qué hacer con todo esto

Timnit Gebru y Émile P. Torres acuñaron el término TESCREAL mientras escribían un artículo juntos. La primera exposición pública es de Torres: «TESCREALism: The Acronym Behind Our Wildest AI Dreams and Nightmares», Truthdig, 15 de junio de 2023. El artículo es «The TESCREAL bundle: Eugenics and the promise of utopia through artificial general intelligence», First Monday 29(4), 14 de abril de 2024.

El acrónimo se despliega como transhumanism, Extropianism, singularitarianism, cosmism (el moderno), Rationalism, Effective Altruism y longtermism. Es decir, nombra exactamente los cinco linajes de este apunte más el racionalismo —el de Yudkowsky, tu lectura 2— y el largoplacismo. La tesis central de Gebru y Torres también es genealógica: rastrean la meta del conjunto hasta el movimiento eugenésico angloamericano, con el transhumanismo como puente.

Hay otras dos críticas que conviene tener a mano. Amia Srinivasan reseñó a MacAskill en la London Review of Books 37(18) del 24 de septiembre de 2015, con el título «Stop the Robot Apocalypse». Su objeción no es aritmética:

Effective altruism doesn't try to understand how power works, except to better align itself with it. In this sense it leaves everything just as it is.

Amia Srinivasan, «Stop the Robot Apocalypse», London Review of Books 37(18), 24 de septiembre de 2015.

Y Torres firma por su cuenta «Why longtermism is the world's most dangerous secular credo», Aeon, 19 de octubre de 2021.

Falta una advertencia, y va contra este apunte. Nombrar un linaje no refuta nada. Una genealogía explica de dónde viene una idea, y de dónde viene una idea no es lo mismo que si es verdadera. Un argumento con antepasados incómodos puede ser correcto y uno con antepasados simpáticos puede ser falso, y eso se decide leyendo el argumento, no el árbol genealógico. Lo que la genealogía sí hace es quitarle a la Declaración el aire de haber caído del cielo y mostrarte de qué discusión es una jugada.

En la sesión vas a discutir estos textos. Hazlo con su versión más fuerte.

Fuente: Escrito para este curso, agosto de 2026, con ayuda de un modelo de lenguaje; las fuentes de cada fragmento van al pie del propio apunte. Material del curso.

Astronomical Waste

Nick Bostrom · 2003

Qué vas a leer. Un artículo de filosofía de 2003, corto: dos mil seiscientas palabras en cuatro secciones, publicado en *Utilitas* 15(3), pp. 308–314. El temario lo cita por esa paginación; aquí se reproduce la versión que el propio Bostrom publica en su sitio, con el mismo texto y sin esos números de página. Es el texto donde el futuro se vuelve una cifra.

Palabras clave. *Desperdicio astronómico*: los recursos del universo que se degradan sin producir vidas mientras no colonizamos. *Costo de oportunidad*: lo que se pierde por retrasar, medido en vidas que pudieron existir. *Riesgo existencial*: el que aniquilaría la vida inteligente originada en la Tierra o recortaría para siempre su potencial. *Vista person-affecting*: la posición según la cual solo cuentan las personas que existen o existirán. *Agregativo y descuento temporal*: las dos condiciones que Bostrom necesita — sumar el bienestar de todos por igual, y no valer menos el futuro por ser futuro.

Qué retener. Tres pasos, y el segundo sorprende. Uno: la aritmética de la sección I. Contando las estrellas del Supercúmulo de Virgo y la capacidad de cómputo de cada una, cada siglo de colonización retrasada cuesta unas 10^{38} vidas humanas potenciales; con el supuesto más conservador, solo cuerpos biológicos, son más de diez billones de vidas por segundo. Dos: la sección III no concluye lo que venía prometiendo. Como las galaxias duran miles de millones de años y los retrasos que podemos causar duran décadas, el riesgo pesa más que la prisa: un punto porcentual menos de riesgo existencial vale más que diez millones de años de demora, y «maximiza la utilidad agregada esperada» se abrevia en «minimiza el riesgo existencial». Tres: la sección IV invierte la recomendación con los mismos números. Si solo cuentan las personas que ya existen, lo urgente vuelve a ser la velocidad, porque hay que llegar vivos. Misma aritmética, distinta población contada, política opuesta.

Es difícil, y está bien. Los exponentes son lo primero que frena, y el propio Bostrom te libera: lo que importa, escribe, no son las cifras exactas sino que son enormes. No intentes imaginarlas. Lo denso de verdad son dos párrafos de la sección IV sobre la utilidad marginal de los recursos y Bill Gates; si te pierdes ahí, el último párrafo de la sección dice la conclusión completa. Las notas al pie van al final y son referencias. Y este artículo reaparece en la lectura 6, que lo cita como su antecedente directo.

I. The Rate of Loss of Potential Lives

As I write these words, suns are illuminating and heating empty rooms, unused energy is being flushed down black holes, and our great common endowment of negentropy is being irreversibly degraded into entropy on a cosmic scale. These are resources that an advanced civilization could have used to create value-structures, such as sentient beings living worthwhile lives.

The rate of this loss boggles the mind. One recent paper speculates, using loose theoretical considerations based on the rate of increase of entropy, that the loss of potential human lives in our own galactic supercluster is at least $\sim 10^{46}$ per century of delayed colonization.¹ This estimate assumes that all the lost entropy could have been used for productive purposes, although no currently known technological mechanisms are even remotely capable of doing that. Since the estimate is meant to be a lower bound, this radically unconservative assumption is undesirable.

We can, however, get a lower bound more straightforwardly by simply counting the number of stars in our galactic supercluster and multiplying this number with the amount of computing power that the resources of each star could be used to generate using technologies for whose feasibility a strong case has already been made. We can then divide this total with the estimated amount of computing power needed to simulate one human life.

As a rough approximation, let us say the Virgo Supercluster contains 10^{13} stars. One estimate of the computing power extractable from a star and with an associated planet-sized computational structure, using advanced molecular nanotechnology², is 10^{42} operations per second.³ A typical estimate of the human brain's processing power is roughly 10^{17} operations per second or less.⁴ Not much more seems to be needed to simulate the relevant parts of the environment in sufficient detail to enable the simulated minds to have experiences indistinguishable from typical current human experiences.⁵ Given these estimates, it follows that the potential for approximately 10^{38} human lives is lost every century that colonization of our local supercluster is delayed; or equivalently, about 10^{29} potential human lives per second.

While this estimate is conservative in that it assumes only computational mechanisms whose implementation has been at least outlined in the literature, it is useful to have an even more conservative estimate that does not assume a non-biological instantiation of the potential persons. Suppose that about 10^{10} biological humans could be sustained around an average star. Then the Virgo Supercluster could contain 10^{23} biological humans. This corresponds to a loss of potential of over 10^{13} potential human lives per second of delayed colonization.

What matters for present purposes is not the exact numbers but the fact that they are huge. Even with the most conservative estimate, assuming a biological implementation of all persons,

the potential for over ten trillion potential human beings is lost for every second of postponement of colonization of our supercluster. 6

II. The Opportunity Cost of Delayed Colonization

From a utilitarian perspective, this huge loss of potential human lives constitutes a correspondingly huge loss of potential value. I am assuming here that the human lives that could have been created would have been worthwhile ones. Since it is commonly supposed that even current human lives are typically worthwhile, this is a weak assumption. Any civilization advanced enough to colonize the local supercluster would likely also have the ability to establish at least the minimally favorable conditions required for future lives to be worth living.

The effect on total value, then, seems greater for actions that accelerate technological development than for practically any other possible action. Advancing technology (or its enabling factors, such as economic productivity) even by such a tiny amount that it leads to colonization of the local supercluster just one second earlier than would otherwise have happened amounts to bringing about more than 10^{29} human lives (or 10^{13} human lives if we use the most conservative lower bound) that would not otherwise have existed. Few other philanthropic causes could hope to match that level of utilitarian payoff.

Utilitarians are not the only ones who should strongly oppose astronomical waste. There are many views about what has value that would concur with the assessment that the current rate of wastage constitutes an enormous loss of potential value. For example, we can take a thicker conception of human welfare than commonly supposed by utilitarians (whether of a hedonistic, experientialist, or desire-satisfactionist bent), such as a conception that locates value also in human flourishing, meaningful relationships, noble character, individual expression, aesthetic appreciation, and so forth. So long as the evaluation function is aggregative (does not count one person's welfare for less just because there are many other persons in existence who also enjoy happy lives) and is not relativized to a particular point in time (no time-discounting), the conclusion will hold.

These conditions can be relaxed further. Even if the welfare function is not perfectly aggregative (perhaps because one component of the good is diversity, the marginal rate of production of which might decline with increasing population size), it can still yield a similar bottom line provided only that at least some significant component of the good is sufficiently aggregative. Similarly, some degree of time-discounting future goods could be accommodated without changing the conclusion. 7

III. The Chief Goal for Utilitarians Should Be to Reduce Existential Risk

In light of the above discussion, it may seem as if a utilitarian ought to focus her efforts on accelerating technological development. The payoff from even a very slight success in this endeavor is so enormous that it dwarfs that of almost any other activity. We appear to have a utilitarian argument for the greatest possible urgency of technological development.

However, the true lesson is a different one. If what we are concerned with is (something like) maximizing the expected number of worthwhile lives that we will create, then in addition to the opportunity cost of delayed colonization, we have to take into account the risk of failure to colonize at all. We might fall victim to an existential risk, one where an adverse outcome would either annihilate Earth-originating intelligent life or permanently and drastically curtail its potential.⁸ Because the lifespan of galaxies is measured in billions of years, whereas the time-scale of any delays that we could realistically affect would rather be measured in years or decades, the consideration of risk trumps the consideration of opportunity cost. For example, a single percentage point of reduction of existential risks would be worth (from a utilitarian expected utility point-of-view) a delay of over 10 million years.

Therefore, if our actions have even the slightest effect on the probability of eventual colonization, this will outweigh their effect on when colonization takes place. For standard utilitarians, priority number one, two, three and four should consequently be to reduce existential risk. The utilitarian imperative “Maximize expected aggregate utility!” can be simplified to the maxim “Minimize existential risk!”.

IV. Implications for Aggregative Person-Affecting Views

The argument above presupposes that our concern is to maximize the total amount of well-being. Suppose instead that we adopt a “person-affecting” version of utilitarianism, according to which our obligations are primarily towards currently existing persons and to those persons that will come to exist.⁹ On such a person-affecting view, human extinction would be bad only because it makes past or ongoing lives worse, not because it constitutes a loss of potential worthwhile lives. What ought someone who embraces this doctrine do? Should he emphasize speed or safety, or something else?

To answer this, we need to consider some further matters. Suppose one thinks that the probability is negligible that any existing person will survive long enough to get to use a significant portion of the accessible astronomical resources, which, as described in opening section of this paper, are gradually going to waste. Then one’s reason for minimizing existential risk is that sudden extinction would off cut an average of, say, 40 years from each of the current (six billion or so) human lives.¹⁰ While this would certainly be a large disaster, it is in the same big ballpark as other ongoing human tragedies, such as world poverty, hunger and disease. On this

assumption, then, a person-affecting utilitarian should regard reducing existential risk as a very important but not completely dominating concern. There would in this case be no easy answer to what he ought to do. Where he ought to focus his efforts would depend on detailed calculations about which area of philanthropic activity he would happen to be best placed to make a contribution to.

Arguably, however, we ought to assign a non-negligible probability to some current people surviving long enough to reap the benefits of a cosmic diaspora. A so-called technological “singularity” might occur in our natural lifetime ¹¹, or there could be a breakthrough in life-extension, brought about, perhaps, as result of machine-phase nanotechnology that would give us unprecedented control over the biochemical processes in our bodies and enable us to halt and reverse the aging process. ¹² Many leading technologists and futurist thinkers give a fairly high probability to these developments happening within the next several decades. ¹³ Even if you are skeptical about their prognostications, you should consider the poor track record of technological forecasting. In view of the well-established unreliability of many such forecasts, it would seem unwarranted to be so confident in one’s prediction that the requisite breakthroughs will not occur in our time as to give the hypothesis that they will a probability of less than, say, 1%.

The expected utility of a 1% chance of realizing an astronomically large good could still be astronomical. But just how good would it be for (some substantial subset of) currently living people to get access to astronomical amounts of resources? The answer is not obvious. On the one hand, one might reflect that in today’s world, the marginal utility for an individual of material resources declines quite rapidly once his basic needs have been met. Bill Gates’ level of well-being does not seem to dramatically exceed that of many a person of much more modest means. On the other hand, advanced technologies of the sorts that would most likely be deployed by the time we could colonize the local supercluster may well provide new ways of converting resources into well-being. In particular, material resources could be used to greatly expand our mental capacities and to indefinitely prolong our subjective lifespan. And it is by no means clear that the marginal utility of extended healthspan and increased mental powers must be sharply declining above some level. If there is no such decline in marginal utility, we have to conclude that the expected utility to current individuals ¹⁴ of successful colonization of our supercluster is astronomically great, and this conclusion holds even if one gives a fairly low probability to that outcome. A long shot it may be, but for an expected utility maximizer, the benefit of living for perhaps billions of subjective years with greatly expanded capacities under fantastically favorable conditions could more than make up for the remote prospects of success.

Now, if these assumptions are made, what follows about how a person-affecting utilitarian should act? Clearly, avoiding existential calamities is important, not just because it would truncate the natural lifespan of six billion or so people, but also – and given the assumptions this is an even weightier consideration – because it would extinguish the chance that current people have of reaping the enormous benefits of eventual colonization. However, by contrast to the total utilitarian, the person-affecting utilitarian would have to balance this goal with another equally important desideratum, namely that of maximizing the chances of current people surviving to benefit from the colonization. For the person-affecting utilitarian, it is not enough that humankind survives to colonize; it is crucial that extant people be saved. This should lead her to emphasize speed of technological development, since the rapid arrival of advanced technology would surely be needed to help current people stay alive until the fruits of colonization could be harvested. If the goal of speed conflicts with the goal of global safety, the total utilitarian should always opt to maximize safety, but the person-affecting utilitarian would have to balance the risk of people dying of old age with the risk of them succumbing in a species-destroying catastrophe. Mixed ethical views, which also incorporate non-utilitarian elements, might or might not yield one of these bottom lines depending on the nature of what is added. 15

M. Cirkovic, “Cosmological Forecast and its Practical Significance”, *Journal of Evolution and Technology*, xii (2002), <https://www.jetpress.org/volume12/CosmologicalForecast.pdf>. ↵

K. E. Drexler, *Nanosystems: Molecular Machinery, Manufacturing, and Computation*, New York, John Wiley & Sons, Inc., 1992. ↵

R. J. Bradbury, “Matrioshka Brains”, Manuscript, 2002, <http://www.aeiveos.com/~bradbury/MatrioshkaBrains/MatrioshkaBrains.html> ↵

N. Bostrom, “How Long Before Superintelligence?”, *International Journal of Futures Studies*, ii (1998); R. Kurzweil, *The Age of Spiritual Machines: When Computers Exceed Human Intelligence*, New York, Viking, 1999. The lower estimate is in H. Moravec, *Robot: Mere Machine to Transcendent Mind*, Oxford, 1999. ↵

N. Bostrom, “Are You Living in a Simulation?”, *Philosophical Quarterly*, liii (211). See also <https://simulation-argument.com>. ↵

The Virgo Supercluster contains only a small part of the colonizable resources in the universe, but it is sufficiently big to make the point. The bigger the region we consider, the less certain we can be that significant parts of it will not have been colonized by a civilization of non-terrestrial origin by the time we could get there. ↵

Utilitarians commonly regard time-discounting as inappropriate in evaluating moral goods (see e.g. R. B. Brandt, *Morality, Utilitarianism, and Rights*, Cambridge, 1992, pp. 23f.). However, it is not clear that utilitarians can avoid compromising on this principle in view of the possibility that our actions could conceivably have consequences for an infinite number of persons (a possibility that we set aside for the purposes of this paper). ↵

N. Bostrom, “Existential Risks: Analyzing Human Extinction Scenarios and Related Hazards”, *Journal of Evolution and Technology*, ix (2002), <https://www.jetpress.org/volume9/risks.html>. ↵

This formulation of the position is not necessarily the best possible one, but it is simple and will serve for the purposes of this paper. ↵

Or whatever the population is likely to be at the time when doomsday would occur. ↵

See e.g. V. Vinge, “The Coming Technological Singularity”, *Whole Earth Review*, Winter issue (1993). ↵

R. A. Freitas Jr., *Nanomedicine*, Vol. 1, Georgetown, Landes Bioscience, 1999. ↵

E.g. Moravec, Kurzweil, and Vinge op. cit.; E. Drexler, *Engines of Creation*, New York, Anchor Books, 1986. ↵

On the person-affecting view, the relevant factor is not the temporal location of a person per se. In principle, effects on the well-being of a past or a future person could also be an appropriate target for moral concern. In practice, however, the effect on the well-being of past persons is likely to be relatively small or even zero (depending on which conception of well-being one adopts), and the effect (of any action that has a significant effect on the future) on not-yet-existing persons is very likely to be such that competing alternative actions would lead to separate sets of possible persons coming into existence, for example as a result of different sperms fertilizing different eggs, and thus there would typically be no future persons whose level of well-being is affected by our current actions. Our actions might affect which future persons there will be, but on the person-affecting view, such results provide no general moral reasons for action. One exception are fetuses and fertilize eggs, which are not yet persons, but which may become persons in the future, and whose lives our current actions may influence. ↵

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The Case for Strong Longtermism

Hilary Greaves y William MacAskill · 2021

Qué vas a leer. Un artículo académico —*working paper* 5-2021 del Global Priorities Institute de Oxford— y la lectura más exigente del módulo. Van aquí **solo las secciones 1 a 4 y la 10**: quedan fuera la 5 (individuos y elección de carrera), la 6 (otras axiologías), la 7 (la objeción de que somos *clueless* sobre el futuro remoto), la 8 (la de fanatismo por probabilidades diminutas), la 9 (la versión deóntica) y el apéndice con la demostración. La sección 10 sí resume todo eso, así que vas a leer el resumen de argumentos que no leíste; sábelo de entrada, y lo mismo con la sección 1, que anuncia secciones que aquí no están.

Palabras clave. *Largoplacismo fuerte*: la tesis de que el efecto sobre el futuro remoto es lo más importante de nuestras acciones de hoy. *ASL*: su versión axiológica precisa, en dos cláusulas — toda opción casi óptima en general lo es para el futuro remoto, y entrega mucho más beneficio allá que acá. *Futuro remoto*: de aquí a cien años en adelante. *Washing-out*: la objeción de que los efectos lejanos se diluyen, que contesta la sección 4. *Estado persistente*: una configuración del mundo de la que, una vez dentro, se tarda muchísimo en salir — la extinción es el caso puro.

Qué retener. Tres cosas. Una: la forma del argumento, que es comparar dos cotas. Arriba, lo mejor que sabemos hacer a corto plazo: repartir mosquiteros contra la malaria, unos cuatro mil dólares por vida salvada, 0.025 vidas por cada cien dólares. Abajo, una cota mínima del beneficio esperado en el futuro remoto. La tesis es que la segunda aplasta a la primera por órdenes de magnitud. Dos: todo corre sobre la *esperanza* del número de seres futuros, no sobre su mediana, y de ahí sale la asimetría que hace el número enorme — creer al 50% que el futuro estará vacío solo divide la cuenta entre dos, mientras que un 1% de crédito a un billón de seres por siglo la multiplica por cien. Tres: la sección 4.3 pone a la inteligencia artificial en el centro, y en la estimación principal de los autores cada cien dólares gastados en seguridad en IA valen tanto como salvar un billón de vidas. El módulo termina aquí porque aquí el aparato de contar produce una instrucción concreta de gasto.

Es difícil, y está bien. Es un paper y se nota, pero lo que más te va a estorbar son las notas al pie. En este recorte caen **entre párrafos**, donde el original cambiaba de página, y a veces parten una oración por la mitad. Cuando un párrafo se interrumpa sin sentido, busca dónde retoma en vez de releerlo. Del resto: lee despacio la sección 2 una vez, porque sus definiciones se usan después al pie de

la letra; en la 3 y la 4 quédate con las cifras y sáltate los análisis de sensibilidad. Y fíjate en el cierre de la 10, donde los propios autores nombran los puntos que consideran más débiles. Es el mejor sitio desde donde empezar a discutirlos.

A striking fact about the history of civilisation is just how early we are in it. There are 5000 years of recorded history behind us, but how many years are still to come? If we merely last as long as the typical mammalian species, we still have over 200,000 years to go (Barnosky et al. 2011); there could be a further one billion years until the Earth is no longer habitable for humans (Wolf and Toon 2015); and trillions of years until the last conventional star formations (Adams and Laughlin 1999:34). Even on the most conservative of these timelines, we have progressed through a tiny fraction of history. If humanity's saga were a novel, we would be on the very first page.

Normally, we pay scant attention to this fact. Political discussions are normally centered around the here and now, focused on the latest scandal or the next election. When a pundit takes a "long-term" view, they talk about the next five or ten years. With the exceptions of climate change and nuclear waste, we essentially never think about how our actions today might influence civilisation hundreds or thousands of years hence.

We believe that this neglect of the very long-run future is a serious mistake. An alternative perspective is given by longtermism, according to which we should be particularly concerned with ensuring that the far future goes well (MacAskill MS). In this article we go further, arguing for strong longtermism: the view that impact on the far future is the most important feature of our actions today. We will defend both axiological and deontic versions of this thesis.

Humanity, today, is like an imprudent teenager. The most important feature of the most important decisions that a teenager makes, like what subject to study at university and how diligently to study, is not the enjoyment they will get in the short term, but how those decisions will affect the rest of their life.

The structure of the paper is as follows. Section 2 sets out more precisely the thesis we will primarily defend: axiological strong longtermism (ASL). This thesis states that, in the most important decision situations facing agents today, (i) every option that is near-best overall is near-best for the far future, and (ii) every option that is near-best overall delivers much larger benefits in the far future than in the near future.

We primarily focus on the decision situation of a society deciding how to spend its resources. We use the cost-effectiveness of antimalarial bednet distribution as an approximate upper bound on attainable near-future benefits per unit of spending. Towards establishing a lower bound on the highest attainable far-future expected benefits, section 3 argues that there is, in expectation, a vast

number of sentient beings in the future of human-originating civilisation. Section 4 then argues, by way of examples involving existential risk, that the project of trying to beneficially influence the course of the far future is sufficiently tractable for ASL(i) and ASL(ii) to be true of the above decision situation. Section 5 argues that the same claims and arguments apply equally to an individual deciding how to spend resources, and an individual

choosing a career. We claim these collectively constitute the most important decision situations facing agents today, so that ASL follows.

The remainder of the paper explores objections and extensions to our argument.

Section 6 argues that the case for ASL is robust to several plausible variations in axiology, concerning risk aversion, priority to the worse off, and population ethics. Section 7 addresses the concern that we are clueless about the very long-run effects of our actions. Section 8 addresses the concern that our argument turns problematically on tiny probabilities of enormous payoffs.

Section 9 turns to deontic strong longtermism. We outline an argument to the effect that, according to any plausible non-consequentialist moral theory, our discussion of ASL also suffices to establish an analogous deontic thesis. Section 10 summarises.

The argument in this paper has some precedent in the literature. Nick Bostrom (2003) has argued that total utilitarianism implies we should maximise the chance that humanity ultimately settles space. Nick Beckstead (2013) argues, from a somewhat broader set of assumptions, that “what matters most” is that we do what’s best for humanity’s long-term trajectory. In this paper, we make the argument for strong longtermism more rigorous, and we show that it follows from a much broader set of empirical, moral and decision-theoretic views. In addition, our argument in favour of deontic strong longtermism is novel.

We believe that strong longtermism is of the utmost importance: that if society came to adopt the views we defend in this paper, much of what we prioritise in the world today would change.

2. Precisifying strong longtermism

2.1 Axiological strong longtermism (ASL)

Strong longtermism could be made precise in a variety of ways. First, since we do not assume consequentialism, we must distinguish between axiological and deontic claims. Let axiological (resp. deontic) strong longtermism be the thesis that far-future effects are the most important determinant of the value of our options (resp. of what we ought to do).

It remains imprecise what “most important determinant” means. Taking the axiological case first, in this paper we consider the following more precise thesis:¹

Axiological strong longtermism (ASL): In the most important decision situations facing agents today,

(i) Every option that is near-best overall is near-best for the far future.

(ii) Every option that is near-best overall delivers much larger benefits in the far future than in the near future.

We discuss deontic strong longtermism in section 9.

Where condition (i) holds, one can identify the near-best options by focussing in the first instance only on far-future effects. If (as we believe, but will not argue here) the analogous statement regarding near-future effects is not also true, that supplies one sense in which far-future effects are “the most important”. Where condition (ii) holds, the evaluation of near-best options is primarily driven by far-future effects. That supplies another such sense.

In sections 3-5, we will argue that clauses (i) and (ii) of ASL hold of particular decision situations: those of a society deciding how to spend money with no restrictions as to ‘cause area’, an individual making the analogous decision, and individual career choice. Because these decision situations have particularly great significance for the well-being of both present and future sentient beings, we claim, they are the most important situations faced by agents today. Therefore, strong longtermism follows, even if ASL(i) and (ii) do not hold of any other decision situations.

Throughout, “the far future” means everything from some time t onwards, where t is a surprisingly long time from the point of decision (say, 100 years). “The near future” means the time from the point of decision until time t . We will interpret both “near-best overall” and “near-best for the far future” in terms of proportional distance from zero benefit to the maximum available benefit, and “much larger” in terms of a multiplicative factor.

As we intend it, ASL is not directly concerned with the objective value of options and their actual effects. Rather, terms like “near-best” and “benefits” relate to the ex ante value of those options, given the information available at the time of decision, and their prospects for affecting the near or far future. Ex ante value may be expected value, but the statement of ASL does not presuppose this.

Since it refers to “benefits”, ASL makes sense only relative to a status quo option: benefits are increases in value relative to the status quo. As above, our primary examples will be cases of deciding how to spend some resource (either money or time). For concreteness, we will then take the status quo to be a situation in which the resources in question are simply wasted. However,

other plausible choices of status quo would be unlikely to significantly affect our argument, and the argument does not require that the status quo be special in any deep sense.

ASL makes only comparative claims. We do not claim, and nor do we believe, that options cannot deliver large benefits without being near-best for the far future, or that available near-future benefits are small in any absolute sense. Our claim is rather that available benefits for the far future are many times larger even than this.

2.2 Benefit ratio (BR) and ASL

Our argument for ASL proceeds via the intermediate claim that the following property holds of the decision situations in question:

Benefit ratio (BR): The highest far-future ex ante benefits that are attainable without net near-future harm are many times greater than the highest attainable near-future ex ante benefits.

We prove in the Appendix that if BR holds of a given decision situation, then (firstly) so does ASL(ii), and (secondly) ASL(i) holds of a certain restriction of that decision situation. (The restriction involves removing any options that do net expected near-future harm; this restriction is innocuous in the context of our argument.)

Evaluating BR, and hence ASL, requires quantitative analysis; any particular quantitative analysis requires strong evaluative assumptions. To this end, we will temporarily make a particular, plausible but controversial, set of evaluative assumptions. Section 6, however, shows that various plausible ways of relaxing these assumptions leave the basic argument intact. One controversial assumption that may be essential, concerning the treatment of very small probabilities, is discussed in section 8.

The inessential assumptions in question include the following. First, we will identify the ex ante value of an option with its expected value: the probability-weighted average of the possible values it might result in. Second, we will identify value with total welfare: that is, we will assume a total utilitarian axiology. Third, and a near-corollary of the latter, we will assume time-separability. The latter allows us to separately define near-future and far-future benefits: overall value is then simply the sum of near-future value and far-future value, where these in turn depend only on near-future (respectively, far-future) effects.

For a rough upper bound on near-future expected benefits in the context of a society spending money, we consider the distribution of long-lasting insecticide-treated bednets in malarial regions, which saves a life on average for around \$4000. Each \$100 therefore saves on average 0.025 lives in the near future (GiveWell 2020a).²

We cannot argue that this is the action with the very largest near-future benefits. In particular, though it seems hard to beat this cost-effectiveness level via any intervention that is backed by rigorous evidence, it might be possible to achieve higher short-term expected benefits via some substantially more speculative route.³ A full examination of the case for strong longtermism would involve investigation of this, and the corresponding sensitivity analysis. However, even quite large upward adjustments to the figure we use here would leave our argument largely unaffected.

We emphasise that we are not considering the long-run knock-on effects of bednet distribution. It is possible, for all we say in this paper, that bednet distribution is the best way of making the far future go well, though we think this unlikely.⁴

Following GiveWell (2018b), we will assume that the short-term benefits of the interventions that do the most short-run good would scale proportionately even if very large amounts of money were spent.

A sister organisation to GiveWell, Open Philanthropy, has tried hard to find human-centric interventions that have more short-term impact, and has struggled (Berger 2019). There might be more cost-effective interventions focused on preventing the suffering of animals living in factory farms (Bollard 2016). We leave this aside in order to avoid getting into issues of inter-species comparisons; again, there is a corresponding need for sensitivity analysis.

It would amount to a “surprising and suspicious convergence” between near-future and far-future optimisation (Lewis 2016).

We will argue in section 4 that, for a society’s decision about how to spend its resources, the lower bound on attainable far-future expected benefits is many times higher than this upper bound for near-future expected benefits, and therefore BR holds of this decision situation. Section 5 discusses related decision situations facing individuals.

3. The size of the future

There is, in expectation, a vast number of lives in the future of human civilisation.⁵ Any estimate of just how “vast” is of course approximate. Nonetheless, we will argue, existing work supports estimates that are sufficiently large for our argument to go through.

There are several techniques one can use for estimating the expected number of future beings. Let us start with the question of the expected duration of humanity’s future existence, temporarily setting aside questions of how large the population might be at any future time.

Firstly, one might use evidence regarding the age of our species to ground judgments on the annual risk of extinction from natural causes, and extrapolate from there. Given that Homo

sapiens has existed for over 200,000 years, Snyder-Beattie et al. (2019:2) thereby estimate that the expected future lifespan of humanity is at least 87,000 years, as far as natural causes of extinction are concerned.

Secondly, one might undertake reference class forecasting (Kahneman and Lovallo 1993; Flyvbjerg 2008). Here, the lifespans of other sufficiently similar species serve as benchmarks. Estimates of the average lifespan of mammalian species (resp. hominins) are between 0.5 and 6 million years (resp. around 1 million years) (Snyder-Beattie et al. 2019:6). Thus reference class forecasting, naively applied, suggests at least 1 million years for the expected future duration of humanity.

Both of these estimates, however, ignore the fact that humans today are highly atypical. Humanity today is significantly better equipped to survive extinction-level threats than either other species are, or than our own species was in the past, thanks to a combination of technological capabilities and geographical diversity. Therefore a range of substantially higher benchmarks is also relevant: for instance, the frequency of mass extinction events (1 in every 30-100 million years (Snyder-Beattie et al. 2019:7)), and the time over which the Earth remains habitable for humans (around 1 billion years (Adams 2008:34)).

The above figures concern the expected duration of humanity's future. Since we are interested in the expected number of future beings, we also need to consider population size.

We again consider several benchmarks. First, the UN Department of Economic and Social Affairs (2019:6) projects that the global population will plateau at around 11 billion people by the year 2100. Second, the large majority of estimates of the Earth's "carrying capacity" – that is, its long-run sustainable population, based on relatively conservative assumptions about future technological progress – are over 5 billion, and sometimes substantially higher

We will use 'human' to refer both to *Homo sapiens* and to whatever descendants with at least comparable moral status we may have, even if those descendants are a different species, and even if they are non-biological.

(Cohen 1998:342; Bergh and Rietveld 2004:197). Third, for predicting the further future, we might extrapolate from the historical trend of human population increasing over time. Given this trend, it is at least plausible that continued technological advances will enable an even larger future population up to some much higher plateau point (say, 1 trillion), even if we cannot currently foresee the concrete details of how that might happen (Simon 1998).

Importantly, it is the expected number of future beings, not the median, that is relevant for our purposes. In addition to the possibility of numbers like the higher benchmarks indicated above, it

is of course also possible that the future duration and/or population size of humanity are much smaller.⁶ However, the effects of these possibilities on the expected number are highly asymmetric. Even a 50% credence that the number of future beings will be zero would decrease the expected number by only a factor of two. In contrast, a credence as small as 1% that the future will contain, for example, 1 trillion beings per century for 100 million years (rather than 10 billion per century for 1 million years) increases the expected number by a factor of 100.

We must also consider two more radical possibilities that, while very uncertain, could greatly increase the duration and future population sizes of humanity. The first concerns space settlement. There are currently no known obstacles to the viability of space settlement, and some scientific investigations suggesting its feasibility using known science (Sandberg and Armstrong 2013; Beckstead 2014). If humanity lives not only on Earth but also on other planets — in our own solar system, elsewhere in the Milky Way, or in other galaxies too — then terrestrial constraints on future population size disappear, and astronomically larger populations become possible. Even if we only settle the solar system, civilisation would have over 5 billion years until the end of the main sequence lifetime of the Sun (Sackmann et al. 1993:462; Schröder and Smith 2008:157-8), and we would have access to over two billion times as much sunlight power as we do now (Stix 2002:6; Sarbu and Sebarchievici 2017:16). If we are able to widely settle the rest of the Milky Way, then we could access well over 250 million rocky habitable-zone planets (Bryson et al. 2021:22), each of which has the potential to support trillions of lives over the course of their sun’s lifetimes. Moreover, an interstellar civilisation could survive until the end of the stelliferous era, on the order of ten trillion years hence (Adams and Laughlin 1999). If we consider possible settlement of the billions of other galaxies accessible to us, the numbers get dramatically larger again.

The second radical possibility is that of digital sentience: that is, conscious artificial intelligence (AI). The leading theories of philosophy of mind support the idea that consciousness is not essentially biological, and could be instantiated digitally (Lewis 1980; Chalmers 1996: ch. 9). And the dramatic progress in computing and AI over just the past 70 years should give us reason to think that if so, digital sentience could well be a reality in the

On duration: technological progress brings not only protection against existing extinction risks, but also novel sources of extinction risk (Ord 2020: esp. chs. 4 and 5). On population size: the tendency for richer societies to have lower fertility rates has led some to conjecture that human population, after plateauing around 2100, might significantly decline into the indefinite future, a high “carrying capacity” notwithstanding (Bricker and Ibbitson 2019).

future. It is also plausible that such beings would have at least comparable moral status to humans (Liao 2020), so that they count for the purposes of the arguments in this paper.⁷

Consideration of digital sentience should increase our estimates of the expected number of future beings considerably, in two ways. First, it makes interstellar travel much easier: it is easier to sustain digital than biological beings during very long-distance space travel (Sandberg 2014:453). Second, digital sentience could dramatically increase the number of beings who could live around one star: digital agents could live in a much wider variety of environments (Sandberg 2014:453), and could more efficiently turn energy into conscious life (Bostrom 2003:309).

One might feel sceptical about these scenarios. But given that there are no known scientific obstacles to them, it would be overconfident to be certain, or near-certain, that space settlement, or digital sentience, will not occur. Imagine that you could peer into the future, and thereby discovered that Earth-originating civilisation has spread across many solar systems. How surprised would you be, compared to how surprised you would be if you won the lottery?

To move towards particular numbers, we consider three specific future scenarios, taken from Newberry (2021a), where civilisation is: (i) Earthbound; (ii) limited to the Solar System; and (iii) expanded across the Milky Way. In each case, Newberry makes a conservative estimate of the carrying capacity of civilisation in that scenario, on the assumptions that digital life is and is not possible, giving six scenarios in all. He also provides a best-guess estimate of the duration of civilisation in that scenario. These scenarios are not meant to exhaust the possibility space, but they give an indication of the potential magnitudes of future population size:

Scenario	Duration (centuries)	Carrying capacity	Number of future (lives per century) lives
Earth (mammalian)	10 ⁴	10 ¹⁰	10 ¹⁴ reference class
Earth (digital life)	10 ⁴	10 ¹⁴	10 ¹⁸
Solar System	10 ⁸	10 ¹⁹	10 ²⁷
Solar System (digital life)	10 ⁷	10 ²³	10 ³⁰
Milky Way	10 ¹¹	10 ²⁵	10 ³⁶
Milky Way (digital life)	10 ¹¹	10 ³⁴	10 ⁴⁵

We return to the likelihood of artificial superintelligence in section 4.3.

To arrive at an overall estimate of the expected number of future people, one would further need to estimate probabilities for scenarios such as those above (and for all other scenarios). However, since the number of lives in the future according to different possible scenarios is spread over many orders of magnitude, in any such expected value calculation, it tends to be the

“largest” scenario in which one has any nonzero credence that drives the overall estimate. Even a 0.01% credence that biological humanity settles the Milky Way at carrying capacity, for example, contributes at least 1032 to the expected number of future beings. Precisely how one’s remaining credence is spread among “smaller” scenarios then makes very little difference.

Because of this, we believe that any reasonable estimate of the expected number of future beings is at least 1024. (In fact, we believe that any reasonable estimate must be substantially higher than this; since higher numbers would make little difference to the arguments of this paper, however, we will not press that case here.) However, we are also sympathetic to the concern that if this is the only estimate we consider, the case for strong longtermism would be driven purely by tiny credences in highly speculative scenarios. We will therefore also consider the extent to which the same arguments would go through on some vastly more conservative estimates, as follows:

Expected number of future beings

Main estimate 1024

Low estimate 1018

Restricted estimate 1014

Our low estimate (1018) corresponds, for instance, to a 0.0000001% credence in the Solar System (biological life) scenario, with zero credence in either digital sentience or more wide-ranging space settlement. Our restricted estimate (1014) corresponds to the above estimate for Earthbound life, with zero credence in any larger-population scenario (including both digital sentience and any space settlement). In the arguments that follow, the reader is invited to substitute her own preferred estimate throughout.

We will argue that BR (and hence ASL) holds of society’s decision situation even on our restricted estimate, and clearly holds by a large margin on our main estimate.

4. Tractability of significantly affecting the far future

The far-future effects of one’s actions are usually harder to predict than their near-future effects. Might it be that the expected instantaneous value differences between available actions decay with time from the point of action, and decay sufficiently fast that in fact the near-future effects tend to be the most important contributor to expected value? If that were so, then neither BR nor ASL would hold.

This is a natural reason to doubt strong longtermism. We will call it the washing-out hypothesis.⁸

We agree that the washing-out hypothesis is true of some decision situations. However, we claim that it is false of our society's decision situation.

Given the argument of section 2, our task is to show that there exists at least one option available to society with the property that its far-future expected benefits are significantly greater than the near-future expected benefits of bednet distribution (that is, recall: 0.025 lives saved per \$100 spent). We will consider examples in two categories: mitigating extinction risk, and positively shaping the development of artificial superintelligence.

4.1 Influencing the choice among persistent states

Here is an abstract structure which, insofar as it is instantiated in the real world, offers a recipe for identifying options whose effects will not wash out.

Consider the space S of all possible fine-grained states the world could be in at a single moment of time (that is, the space of all possible instantaneous microstates). One can picture the history of the universe as a path through this space. Let a persistent state be a subset A of S with the property that, given the dynamics of the universe, if the instantaneous state of the world is in A , then the expected time for which it remains in A is extremely long. Now suppose that there are two or more such persistent states, differing significantly from one another in value. Suppose further that the world is not yet in any of the states in question, but might settle into one or the other of the states in question in the foreseeable future. Finally, suppose that there is something we can do now that changes the probability that the world ends up in a better rather than a worse persistent state. Then, as a result of the persistence that is built into the definition, the effects of these actions would not wash out at all quickly.

The empirical question is whether there are, in the real world, any options available that instantiate the structure just described. We claim that there are.

4.2 Mitigating risks of premature human extinction

The non-existence of humanity is a persistent state par excellence. To state the obvious: the chances of humanity re-evolving, if we go extinct, are miniscule. Only slightly more subtly, the existence of humanity is also a persistent state: while we face significant risks of premature extinction, as argued in section 3, humanity's expected persistence is vast.

These persistent states have unequal expected value. Assuming that on average people have lives of significantly positive welfare,⁹ according to total utilitarianism the existence of humanity is significantly better than its non-existence, at any given time. Combining this with

It is important here to distinguish between ex ante and ex post versions of the washing-out claim. The ex post version is false, as is established by the literature on cluelessness; cf. section

7.1. However, it is the much more plausible *ex ante* washing-out claim that is relevant to the arguments of this paper.

We return to this assumption in section 6.

the fact that both states are persistent, premature human extinction would be astronomically bad. Correspondingly, even an extremely small reduction in extinction risk would have very high expected value (Bostrom 2013:18). For example, even if there are ‘only’ 10¹⁴ lives to come (as on our restricted estimate), a reduction in near-term risk of extinction by one millionth of one percentage point would be equivalent in value to a million lives saved; on our main estimate of 10²⁴ expected future lives, this becomes ten quadrillion (10¹⁶) lives saved.

As is increasingly recognised, as an empirical matter of fact, there are things we could do that would reduce the chance of premature human extinction by a non-negligible amount. As a result, although precise estimates of the relevant numbers are difficult, the far-future benefits of some such interventions seem to compare very favourably, by total utilitarian lights, to the highest available near-future benefits.

The detection and potential deflection of asteroids provides a relatively robust example of such an intervention. This involves scanning the skies to identify asteroids that could potentially collide with Earth and, if one were found, investing the resources to try to deflect it, and/or to prepare bunkers and food stockpiles to help us survive an impact winter. Most of the expected costs here are in detection, because the costs of deflection and preparation are only paid in the very unlikely event that one does detect an asteroid on a collision course with Earth.

In 1996, NASA commenced the Spaceguard Survey, a multi-decade plan to track near-Earth objects with the aim of identifying any on impact trajectories. At a total cost of \$71 million (USD) by 2012, the Spaceguard Survey had tracked over 90% of asteroids of diameter 1km or more in near-Earth orbit, and all asteroids of diameter 10km or more over 99% of the sky.

It is not certain that a large asteroid collision would cause human beings to go extinct. We assume a status quo risk of human extinction, conditional on the impact of a 10km+ asteroid, of 1%. It is also far from certain that we could deflect a 10km+ asteroid, even if we knew it was on a collision course. However, it is far from certain that we could not, and, as above, there are other actions we could take to protect against the extinction risk. We assume here that if such an object were detected to be on a collision course, our deflection and preparation efforts would reduce extinction risk by a proportional 5%. The assumptions in this paragraph follow Newberry (2021b), and seem fairly conservative.

Putting these numbers together, we estimate that the Spaceguard Survey, on average, reduced extinction risk by at least 5×10^{-16} per \$100 spent. On our main estimate of the expected number of future beings, this amounts to 500 additional million lives; this decreases to 500 or 0.05 lives on our low and restricted estimates, respectively.

Of course, we should expect further work on asteroids to have lower cost-effectiveness, because of diminishing marginal returns. However, the opportunity remains significant. The remaining risk of a 10km+ asteroid collision in the next 100 years has been estimated at 1 in 150 million (Ord 2020:71). It has been estimated that the cost to detect with near-certainty

any remaining asteroids of greater than 10km diameter would be at most a further \$1.2 billion (Newberry 2021b). On our main (resp. low, restricted) estimate of the expected number of future beings, every \$100 of this would, on average, result in 300,000 (resp. 0.3, 0.00003) additional lives. This example therefore supports strong longtermism on our main and low estimates, though not on the restricted estimate. Organisations whose work mitigates risk of extinction from asteroid impacts, and which would benefit from substantially more funding, include the Planetary Society and the B612 Foundation.

While asteroid defense is among the more easily quantified areas of extinction risk reduction, it is far from the only one, or the most significant (Ord 2020: ch. 3). Another possibility concerns global pandemics. Such a pandemic could be natural or man-made, with the latter being particularly concerning (Posner 2004:75-84; Rees 2018: sec. 2.1; Ord 2020). In particular, progress in synthetic biology is very rapid (Meng and Ellis 2020), and it is likely that we will soon be able to design man-made viruses with very high contagiousness and lethality. If such pathogens were released (whether deliberately or by accident (Shulman 2020; Ord 2020:129–131)) in the course of military tensions, or by a terrorist group, there is a real possibility that they could kill a sufficient number of people that the human species would not recover.

In a recent paper, Millet and Snyder-Beattie (2017) use three distinct methods to generate estimates of the risk of an extinction-level pandemic in the next 100 years. The resulting estimates range from 1 in 600,000 to 1 in 50. The authors further use figures from the World Bank to generate a very conservative estimate that \$250 billion of spending on strengthening healthcare systems would reduce the chance of such extinction-level pandemics this coming century by at least a proportional 1%.¹⁰

Taking the geometric mean to average across the two methods that generate the lower estimates for extinction risk, we obtain a risk of about 1 in 22,000 of extinction from a pandemic over the next 100 years.¹¹ If we use the above figure of \$250 billion to reduce the risk by 1%, and assume that the risk reduction occurs throughout the next 100 years but only in that time period,

then each \$100 of such spending would, in expectation, increase the number of future beings by 200 million (respectively, 200, 0.02) on our main (resp., low, restricted) estimate. According to these calculations, the far-future benefits would thereby significantly exceed the near-future benefits of bednet distribution on our main and low estimates of the size of the future, though not on our restricted estimate. Organisations

Two ways in which Millet and Snyder-Beattie's estimate is particularly conservative are (i) that the \$250bn figure is at the extreme upper end of anticipated costs for the intervention they discuss, and (ii) that the intervention in question concerns an extremely broad-based approach to biosecurity, not specifically optimising for extinction risk reduction.

We use the geometric rather than the arithmetic mean because the estimates in question are spread across several orders of magnitude; the arithmetic mean effectively defers to the highest estimate on the question of order of magnitude. Using the arithmetic mean would lead to results that are still more favourable to strong longtermism. Similarly, we disregard Millet and Snyder-Beattie's "Model 1" because, as the authors note, this model is flawed in important respects; including this model would also strengthen the case for strong longtermism.

working on these threats include the John Hopkins Center for Health Security, the Nuclear Threat Initiative's biosecurity program, and Gryphon Scientific.

4.3 Influencing the choice among non-extinction persistent states

A second way of positively impacting the long run is by improving the value of the future conditional on the existence of a very large number of future sentient beings. For concreteness, we focus on one way of doing this: positively shaping the development of artificial superintelligence (ASI), that is, artificial systems that greatly exceed the cognitive performance of humans in virtually all domains of interest.¹²

The idea that the development of sufficiently advanced artificial intelligence could prove a key turning point in history goes back to the early computer pioneers Alan Turing (1951) and I.J. Good (1966). It has more recently been the subject of wider concern.¹³ There are two classes of long-term worry.

The first is from AI-takeover scenarios (Bostrom 2014; Russell 2019). This worry is that, once we build a human-level artificial intelligence, it would be able to recursively self-improve, designing ever-better versions of itself, quickly becoming superintelligent. From there, in order to better achieve its aims, it will try to gain resources, and try to prevent threats to its survival. It would therefore be incentivised to take over the world and eliminate or permanently suppress

human beings. Because the ASI's capability is so much greater than that of humans, it would probably succeed in these aims.

The second worry is from entrenchment scenarios (MacAskill MS). If an authoritarian country were the first to develop ASI, with a sufficient lead, they could use this technological advantage to achieve world domination. The authoritarian leader could then quash any ideological competition. An AI police force could guarantee that potential rebellions are prevented; an AI army would remove any possibility of a coup. And if the leader wanted his ideology to persist indefinitely, he could pass control of society on to an ASI successor before his death. To this end, he could hard-code the goals of the ASI to match his own, have the ASI learn his goals from his speech and behaviour, or even 'mind upload', scanning his brain and having it digitally emulated (Sandberg and Bostrom 2008; Sandberg 2013).

In either of these scenarios, once power over civilisation is in the hands of an ASI, this could persist as long as civilisation does (Riedel MS). Different versions of the ASI-controlled futures are therefore persistent states with significantly differing expected value, so that we have another instantiation of the structure outlined in section 4.1. The ruler-ASI could monitor every aspect of society. And it could replicate itself indefinitely, just as easily as we can replicate software today; it would be immortal, freed from the biological process of

Other areas one might consider here include affecting the values that the world converges on (Reese 2018), or reducing the risk of a totalitarian world government (Caplan 2008).

Those concerned include leading machine learning researchers such as Stuart Russell (2019) and Shane Legg (2008: sec. 7.3), philosophers such as Nick Bostrom (2014), Eliezer Yudkowsky (2013), Toby Ord (2020:138–152) and Richard Ngo (2020), physicists such as Max Tegmark (2017: ch. 4) and Stephen Hawking (2018: ch. 9), and tech entrepreneurs such as Elon Musk (2014), Sam Altman (2015) and Bill Gates (Statt 2015).

aging. The value of the resulting world would depend in considerable part on the goals of the ruler-ASI.

Though extinction risks involve dramatic reductions in the size of the future population, these AI scenarios need not. In the classic statement of the AI-takeover scenario, the ASI goes on to settle the stars in pursuit of its goals (Bostrom 2014:100). Similarly, if an authoritarian leader transferred power to an ASI, they too might want their civilisation to be large, populous and long-lasting. In particular, for a wide variety of goals (such as building the grandest possible temples, doing research, or, in a toy example Bostrom (2014:123-4) gives to illustrate the general phenomenon of misaligned AI, maximising the number of paperclips), acquiring more resources

helps with achievement of these goals, which motivates settling the stars. And, in order to fulfill these goals, a populous workforce would be instrumentally valuable. In expectation, the number of future beings, in these scenarios, is very large.

Now, this workforce might consist almost entirely of AIs. But, as we noted in section 3, there are reasons to think that such beings would have moral status, and therefore how well or poorly their lives went would be of moral concern, relevant to the arguments of this paper. And, at least on the authoritarian-takeover scenarios, the ruler might wish to have a very large number of human followers, too.

There are two strands of work aimed at reducing risks from ASI. First, AI safety research, which aims to ensure that AI systems do what we intend them to do (Amodei et al. 2016). Such work is conducted by organisations such as Berkeley’s Center for Human-Compatible AI, the Machine Intelligence Research Institute, and labs within Google DeepMind and OpenAI. Second, policy work, in particular to ensure a cooperative approach between countries and companies: for example, by The Partnership on AI, the Centre for the Governance of AI, and the Center for New American Security.

Despite this work, ASI safety and policy are still extremely neglected. For example, the Open Philanthropy Project is the only major foundation with these issues as a key focus area; it spends under \$30 million per year on them (Open Philanthropy 2020).¹⁴ The AI safety teams at OpenAI and DeepMind are small.

There is no hard quantitative evidence to guide cost-effectiveness estimates for AI safety work. Expert judgment, however, tends to put the probability of existential catastrophe from ASI at 1-10%.¹⁵ Given these survey results and the arguments we have canvassed, we think

Neglectedness is crucial to the argument of this paper. Would strong longtermism still be true if, for example, 10% of global GDP were already spent on the most valuable long-term-oriented interventions? Even if true, would it still be significantly revisionary compared to a near-termist approach, as we have claimed it is at the current margin? We aren’t sure. Our claim here is only that the world today is clearly far below this optimum.

Grace et al. (2018) asked 352 leading AI researchers to give a probability on the size of existential risk arising from the development of ‘human-level machine intelligence’; the median estimate was 5%. A survey among participants at a conference on global catastrophic risks similarly found the median estimate to be 5% (Sandberg and Bostrom 2008). One would expect a selection effect to be at work in surveys of those who have chosen to work on existential risk, but not so (or not strongly) for the survey of AI researchers.

that even a highly conservative assessment would assign at least a 0.1% chance to an AI-driven catastrophe (as bad as or worse than human extinction) over the coming century. We also estimate that \$1 billion of carefully targeted spending would suffice to avoid catastrophic outcomes in (at the very least) 1% of the scenarios where they would otherwise occur. On these estimates, \$1 billion of spending would provide at least a 0.001% absolute reduction in existential risk. That would mean that every \$100 spent had, on average, an impact as valuable as saving one trillion (resp., one million, 100) lives on our main (resp. low, restricted) estimate – far more than the near-future benefits of bednet distribution.

4.4 Uncertainty and ‘meta’ options

There is a lot of uncertainty in the numbers we have given, even in the most scientifically robust case of asteroid detection. We will give this issue a more thorough treatment in the next section, arguing against various ways in which one might worry it undermines our argument.

One thing that uncertainty can support, however, is a preference for different types of strategy to improve the far future. Rather than directly trying to influence the far future, one could instead try to invest in decision-relevant research, or invest one’s resources for use at a later date.

The possibility of either of these strategies strengthens our argument considerably. To see this, let us suppose, for the sake of argument, that no ‘first-order’ intervention (such as those we discussed in sections 4.2-3) delivers higher far-future expected benefits than the highest available near-future expected benefits, relative to the credences that are appropriate in the present state of information. Suppose, however, that it is highly likely that conditional on sufficient additional information, at least one of the proposed interventions, or another such intervention (not yet considered) in a similar spirit, would have much higher far-future benefits, relative to the updated credences, than the best available near-future benefits. Then society might fund research into the cost-effectiveness of various possible attempts to influence the far future. Provided that subsequent governments or philanthropists would take due note of the results, this ‘meta-option’ could easily have much greater far-future expected benefits than the best available near-future expected benefits, since it could dramatically increase the expected effectiveness of future governmental and philanthropic action (all relative to currently appropriate credences).

A complementary possibility is that rather than spending now, society could save its money for a later time (Christiano 2014; MacAskill 2019; Trammell 2020). That is, it could set up a sovereign wealth fund, with a longtermist mission. This fund would pay out whenever there becomes available some action that will sufficiently benefit the far future (in expectation), whether that is during the lifetimes of current citizens or later. There would be some annual risk of future governments being misaligned and using the money poorly, but this risk could

be mitigated via constitutional enshrinement of the mission, and would be compensated by the fact that the fund would benefit from compound returns of investment.¹⁶

These considerations show that the bar that ‘intractability’ objections to our argument must meet is very high. For BR to fail to hold on such grounds, every option available to society must have negligible effect on the far future’s expected value. Moreover, it must be near-certain that there will be no such actions in the future, and that no such actions could be discovered through further research. This constellation of conditions seems unlikely.

10. Summary and conclusions

The potential future of civilisation is vast. Once we appreciate this, it becomes plausible that impact on the far future is the most important feature of our actions today.

Strong longtermism would be false in a world that had sufficiently weak causal connections between the near and the distant future, such that it was too difficult to significantly influence

There are, however, reasons to think that these sacrifices are not as great as we might initially suppose (MacAskill, Mogensen and Ord 2018).

Mogensen (2020) discusses specifically the relationship between demandingness and longtermism.

Might our arguments go further than this, and justify atrocities in the name of the long-term good? Perhaps the French Revolution had good long-term consequences, in terms of bringing about a more liberal and democratic world: does strong longtermism, if so, justify the guillotine? We do not think so, for at least two reasons. The first is that, for such serious side-constraints, something closer to absolutism or near-absolutism becomes much more plausible (or, at least, it takes more than mere ex ante goodness to justify violation of those side-constraints). The second is that, in almost all cases, when there is some option available that promotes the long-term good while violating a serious side-constraint, there will be some alternative option available that achieves a similar amount of long-term good without violating that side-constraint. Liberal democracy could have been achieved in France without the Reign of Terror.

the course of the very long-run future. However, we have argued, the world we find ourselves in today does not have this feature.

We presented our central case in terms of (i) a total utilitarian axiology and (ii) an expected value treatment of decision-making under uncertainty. However, we argued, plausible deviations from either or both of these assumptions do not undermine the core argument.

This paper mainly focussed on the decision situations of a society or individual considering how to spend money without constraints as to cause area, and of an individual's career choice. We argued that these are situations where we can in expectation significantly influence the far future. Precisely because of this, they are among the most important decision situations we face, and axiological strong longtermism follows.

In our own view, the weakest points in the case for axiological strong longtermism are the assessment of numbers for the cost-effectiveness of particular attempts to benefit the far future, the appropriate treatment of cluelessness, and the question of whether an expected value approach to uncertainty is too “fanatical” in this context. These issues in particular would benefit from further research.

In addition to axiological issues, we also discussed the counterpart deontic issues. We suggested that deontic strong longtermism might well be true even if consequentialism is false, on the grounds that (i) the stakes involved are very high, (ii) a plausible non-consequentialist theory has to be sensitive to the axiological stakes, becoming more consequentialist in output as the axiological stakes get higher, and (iii) in the key decision situations, any countervailing constraints and/or prerogatives are comparatively minor. Quite plausibly, in the world as it is today, the most important determinants of what we ought to do arise from our opportunities to affect the far future.

It is possible, but far from obvious, that far-future impacts are also more important than near-future impacts in a much wider class of decision situations: for instance, decisions about whether or not to have a child, and government policy decisions within a relatively narrow ‘cause area’. Insofar as they are, strong longtermism could potentially set a methodology for further work in applied ethics and applied political philosophy: for each issue in these subfields, one could identify the potential far-future effects from different actions or policies, and then work through how these bear on the issue in question. The answers might sometimes be surprisingly revisionary.

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